

A project report
on
**EXPERIMENTAL STUDY OF NANO-PCM
TRIPLE TUBE THERMAL ENERGY
STORAGE SYSTEM**

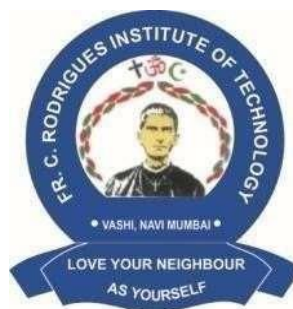
Submitted in partial fulfillment of the requirements
of the degree of

Bachelor of Engineering
In
Mechanical Engineering
By

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2020-2021

Certificate

This is to certify that the project entitled “**Experimental study of Nano-PCM triple tube thermal energy storage system**” is a bonafide work of “**Noel Baby**” (201741), “**Mansi Patil**” (201745), “**Ritik Florian**” (201752), “**Meet Shah**” (201772) submitted to the University of Mumbai in partial fulfilment of the requirement for the award of the degree of “**Bachelor of Engineering**” in “**Mechanical Engineering**”.

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Abstract

The sources of fossil fuels are limited and are becoming too expensive or too environmentally damaging to retrieve. The TES (Thermal Energy Storage) is considered as a promising technology to meet future energy demands. Thermal energy storage based on PCM (Phase Change Materials) as energy storage material is beneficial due to their low cost and high storage capacity at the isothermal condition. The thermal conductivity of PCM for thermal energy storage, is relatively weak, which is one of the main drawbacks for limiting their utility. In the recent past, few attempts have been made to enhance the thermal conductivity of PCM by mixing different additives in the appropriate amount. NePCM (Nano-enhanced Phase Change Material) improves the thermal performance of base PCM by enhancing heat transfer rate during storage. The addition of nanomaterial to the PCM enhances its thermal conductivity, hence reducing the charging and discharging time. The triple tube heat exchanger consists of an annulus shaped PCM layer that is in contact with the HTF (Heat Transfer Fluid) from both inner and outer surfaces provides a larger heat transfer area. Hence, in the present study, a comparison between the system with PCM and system with nano-enhanced PCM in triple tube heat exchanger will be tested for their amount of heat transfer rate.

Keywords: *Thermal energy storage, PCM, Nanomaterials.*

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List of Abbreviations and Notations

PCM	Phase Change Material
NePCM	Nano-enhanced Phase Change Material
HTF	Heat Transfer Fluid
Φ	Amount of Nanomaterial
ρ_{pcm}	Density of Phase Change Material
ρ_{npcm}	Density of Nano-enhanced Phase Change Material
ρ_{np}	Density of Graphene Nanomaterial
C_{ppcm}	Specific Heat of Phase Change Material
C_{pnpcm}	Specific Heat of Nano-enhanced Phase Change Material
C_{pg}	Specific Heat of Graphene Nanomaterial
K_{pcm}	Thermal Conductivity of Phase Change Material
K_{npcm}	Thermal Conductivity of Nano-enhanced Phase Change Material
K_{g}	Thermal Conductivity of Graphene Nanomaterial
μ_{pcm}	Dynamic viscosity of Phase Change Material
μ_{npcm}	Dynamic viscosity Nano-enhanced Phase Change Material
L_{pcm}	Latent Heat of Phase Change Material
L_{npcm}	Latent Heat of Nano-enhanced Phase Change Material
L.F	Liquid Fraction

Chapter 1

Introduction

Thermal energy storage is a technology that stores thermal energy by heating or cooling a storage medium so that the stored energy can be used afterwards for heating and cooling applications and power generation. The use of an energy storage system can considerably decrease the gap between supply and demand of energy.

1.1 Background

The demand for energy for sustaining is on a steep increment due to the ever-increasing extent of the world's population. The supply of energy is constantly under stress to provide the needs of every living being on this planet. Today, when the fossil fuels are depleting day by day, the use of renewable energy is on high demand. However, there has always been a major

fluctuation in the demand and supply relation of energy. To get rid of this, there is a necessity for storing the available energy as much as we can. Hence, engineering disciplines are taking efforts to store the thermal energy to the most possible extent to contribute to the sustainable future of this planet.

1.2 Motivation

In the present age, there is an immense need for storing heat energy due to the continuous fluctuating demand and supply of energy. PCM plays one such role in harnessing, storing and reusing the heat energy. The intermittent availability of solar energy problem can be solved by thermal energy storage. There is a need for designing a simplified and effective model of the heat exchanger to store the energy. Currently, PCM based thermal energy storage system along with nanoparticles is widely under experimentation worldwide. The number of papers has shown a significant increase in the thermal conductivity of PCM just by adding nanoparticles. These are the aspects that provided us with the motivation to carry out the research activities on this topic.

1.3 Aim

To design and fabricate PCM based thermal energy storage system.

1.4 Objectives

The objectives of the present study are -

- To design and fabricate a thermal energy storage system with active or passive techniques.
- To enhance the rate of heat transfer in thermal energy storage system.

1.5 Literature Review

Thermal energy storage is a technology that encourages the rapidly decreasing energy to be stored in a storage system using PCM that allow energy storage using its invaluable thermal properties that can be later used for various applications in society. Amrouche et al.^[1] studied in detail the advantages and disadvantages of various thermal storage systems and listed the benefits of Latent heat system for heat storage and application is studied. Eslami M. and Bahrami^[2] studied that Thermal energy can be stored as sensible heat in heat storage media, as latent heat associated with phase change materials (PCMs) or as thermo-chemical energy associated with chemical reactions. Also concluded the problem of sensible and latent thermal energy storage tank is studied in a PCM latent storage, conduction is a dominant process at a earlier times of charging process. R.Sudheer and K.N.Prabhu^[3] studied the cooling curve analysis of thermal energy storage material using PCMs, metallic PCMs showed higher cooling rates and thermal diffusivity, with the use of advanced data acquisition systems along with high sensitive thermocouples the thermo physical properties such as thermal conductivity and density results will be more accurate. B. Zalba et al.^[4], Abdul Jabbar N,et al.^[5] and Jiang Zhao et al.^[6] concluded that PCM allows a colossal quantity of energy to be stored in relatively small volumes, resulting in one of the lowest storage media costs of any storage concepts. A. Sharma et al.^[7] reviewed the thermal energy storage technology with PCMs of different applications and assesses the thermal properties of various PCMs classified as organic, inorganic and eutectic PCMs. Organic PCMs constituted by a wide range of materials including paraffin, fatty acids and their eutectic mixtures, esters and other organic compounds whereas the inorganic materials are hydrated salts and metallics. The organic materials are more chemically stable than inorganic substances, they melt congruently, and supercooling does not pose a significant problem. V.V. Tyagi and D. Buddhi^[8] concluded a test where they found out good stability of thermal properties i.e. latent heat of fusion and melting temperature and both properties showed only slight variation in nature. R K. Sharma, et al.^[9] experimented the properties of paraffin after 1500 thermal cycle test using a DSC where it showed thermal and chemical stability, therefore having considerable advantage over other organic PCMs.

1.6 Summary

From the review of above-mentioned journal papers, it can be concluded that thermal energy storage systems can be designed to store sensible heat as well as the latent heat between which the latent heat storage is the most suitable and efficient way. Organic PCMs are better than Inorganic ones for achieving an effective performance of thermal energy storage system. Paraffin wax OM50 can be proved as the most compatible material for storing thermal energy in relatively small volumes with one of the lowest costs among any storage media. Water is chosen as it is the better Heat Transfer Fluid.

1.7 Problem definition

Design and fabrication of NePCM based triple tube thermal energy storage system.

1.8 Scope of the project

The scope of the present study is

- To design and fabricate triple tube heat exchanger based thermal energy storage system.
- Taking OM-50 as the PCM and graphene as the nanomaterial.
- The operating temperature of the HTF will remain around 55°C - 60°C.
- To calculate and compare the melting time, solidification time and efficiency of the two systems.

Chapter 2

Literature review

Many scientists and Engineers worldwide are currently carrying out thorough studies of the PCM based thermal energy storage systems enriched with Nano particles. A detailed study of number of journal papers was performed and based on the conclusion of each of them, the selection of various required parameters for the current project were made. These are discussed in the following section.

2.1 Literature Review

Most PCMs have a limitation of very low thermal conductivity that leads to the poor performance of the PCM-based thermal energy storage systems. Nanoparticles have made a revelation in all fields of engineering and technology. Materials of size less than 100 nm are

termed as nanoparticles. Due to the excellent properties shown by nanoparticles, they have been used for various applications such as heat transfer, environmental, medical, catalysis, and imaging applications. The different types of nanoparticles widely used for the various applications include metal nanoparticles, ceramic nanoparticles, polymeric nanoparticles. One of the most widely used applications of nanoparticles is the thermal conductivity enhancement of conventional heat transfer fluids. Boda, M. A. et al.^[10] studied that most organic PCMs are chemically stable and non-corrosive. The PCMs with an easily adjustable melting point would be a necessity as the melting point the most important criterion for the selecting a PCM for passive applications. Bose, et al.^[11] reviewed the various performance enhancement techniques on LHESS with only paraffin wax as PCM. A comprehensive review on synthesis, characterization and property of NePCM with paraffin wax as the base PCM has been made. This paper has also base Paraffin wax as the cheapest and most widely available PCM with many other desirable characteristics making them a potential material for low and medium temperature heat storage applications. Bondareva, et al.^[12] studied that the addition of a small amount of nanoparticles allows the reduction of the source temperature, increasing the heat transfer rate inside the PCM; however, at higher nanoparticle concentrations (3% or more), an increase in the melt viscosity reduces the natural circulation caused by buoyancy forces, and an improvement of the thermal conductivity in the material does not compensate for this effect and also shown that the effect of nanoparticles on system performance depends on the intensity of convective heat transfer. Nematpour Keshteli, et.al^[13] explained the outputs that showed the existence of PCMs and NEPCMs could have a positive impact on enhancing the energy storage and building performance. Dispersing nanoparticles in the PCM augments the thermal conductivity more than pure PCM. Fan, et al.^[14] showed the planar GNPs, however, outperform significantly all the counterparts due to their geometry-induced low thermal interface resistance. The greatest thermal conductivity enhancement of GNP-based nanocomposite PCMs has been measured to be nearly 170% at the loading of 5 wt.%, the GNPs have been shown to be a promising filler material for preparing nanocomposite PCMs with greatly enhanced thermal conductivity and moderately decreased energy storage capacity, which have great potential in TES applications. Kant, K., Shukla,.et.al^[15] has experimented the melting of three different PCM filled in an aluminum square cavity subjected to a constant vertical wall temperature. The effect of emulsifying the graphene nanoparticles in PCM results in a relative increase of the dynamic viscosity compared with that of the pure PCM, thus considerably degrading its natural convection heat transfer efficiency with the increased concentration of nanoparticles across the melted region. They also said that the effective thermal conductivity of all three latent heat storage media can be significantly increased by using smaller volumetric concentrations of

graphene particles although the convection heat transfer gets hampered by the same additives. Le Ba, et.al.^[16] found that two step preparation method has been accepted by a majority of researchers. It also reviews that GNF and has investigated its properties pivoting around the shape of graphene nanoparticles. Further important investigation reveal long term stability for many practical applications. Başal, B., and Ünal, A.^[17] studied the performances of triple tube and conventional double tube heat exchanger have been studied to conclude the fact that the melting time can be reduced significantly in triple tube configuration and also found that in triple tube heat exchanger the radial position and the thickness of PCM layer are the most important parameters that affect charging time and capacity of storage. Kadivar, et.al^[18] showed the present study has considered optimizing the performance of a TES system, in which a phase-change material (PCM) is placed in the eccentric annulus of a horizontal double-pipe heat exchanger, by varying the annulus radial and tangential eccentricities of this configuration.

2.2 Summary

The literature review performed with the above-mentioned journal papers concluded that nanoparticle enhanced PCM based thermal energy storage system can prove more efficient than the regular PCM based system. We also concluded that organic PCMs work better than the inorganic ones wherein OM50 to be the most suitable to use. While selecting the nanoparticles graphene has proved the best and most compatible among many other nano materials. Water is the most appropriate option for taking as a HTF as it is readily available and non-reactive in nature. The thermal conductivity of PCM is significantly increased due to the insertion of nanoparticles and hence can be availed as a better alternative for enhancing heat storage applications. Triple tube heat exchanger is the platform for the experiment considered for experimental investigation.

Chapter 3

Theoretical Analysis

A detailed theoretical study was conducted after referring to the number of research papers and various literature material for the purpose of selecting the most appropriate PCM and the most compatible Nanoparticle. Various methods for synthesizing Nanofluid were also studied and the best method was chosen. The further part gives a detailed explanation of the theoretical analysis performed.

3.1 PCM selection

PCM as the name suggests is a material which changes its phase when a significant heat is provided. It is the most critical and significant part of the present study. Hence various aspects were taken into consideration while selecting the most suitable PCM which are briefed further.

PCMs are “Latent” heat storage materials. The thermal energy transfer occurs when material changes from solid to liquid, or liquid to solid as shown in the fig 3.1. This is called a change in state or “Phase”. Initially, these solid-liquid PCMs perform like conventional storage materials, their temperature rises as they absorb heat. Unlike conventional (sensible) storage materials, PCMs absorb and release heat at a nearly constant temperature. They store 5–14 times more heat per unit volume than sensible storage materials such as water, masonry, or rock. A large number of PCMs are known to melt with a heat of fusion in any required range. However, for their employment as latent heat storage materials, these materials must exhibit certain desirable thermodynamic, kinetic and chemical properties.

PCMs are currently attracting a lot of attention for the heat-storing application due to the progressive reduction in the cost of renewable electricity, coupled with limited hours of availability, resulting in a misfit between peak demand and availability of supply.

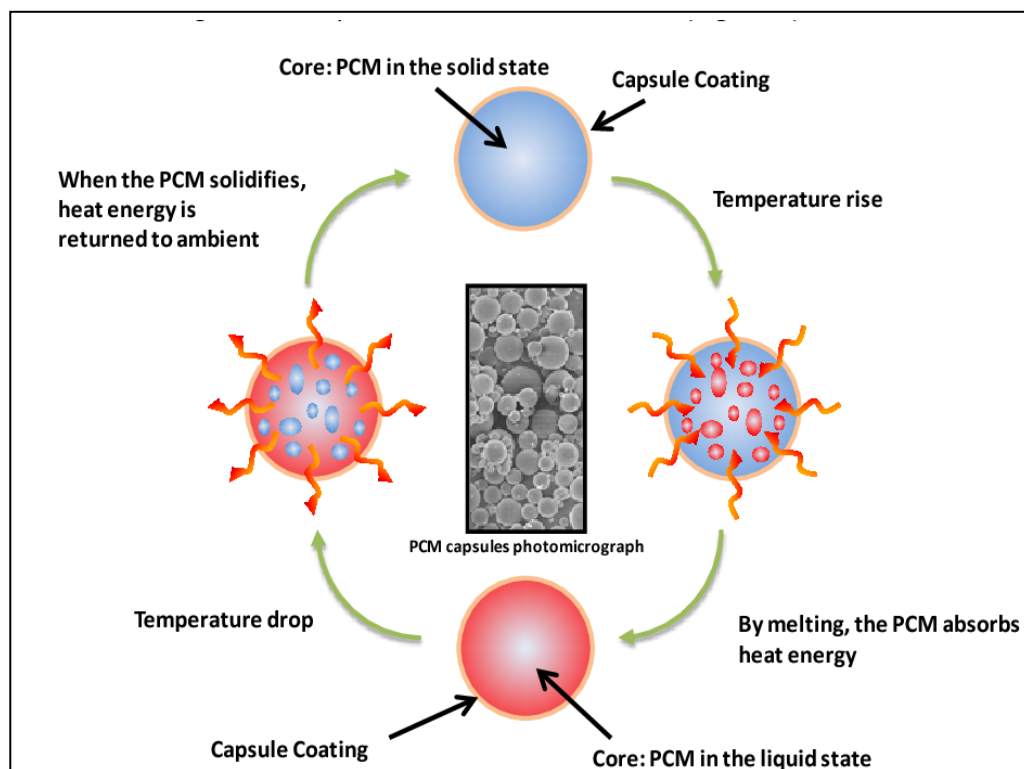


Fig 3.1. Working of PCM

3.2 Classification of PCM

PCMs are mainly grouped into 3 types- Organic, Inorganic and Eutectic as shown in fig. 3.2. The inorganic compounds have the highest latent heat, but these materials are disposed to under- cooling, so the phase changing do not come off in the melting point in every case. The latent heat cannot be used if the material stays in the liquid phase during the discharging. Hence it is not desirable to choose this category as our aim is to utilize the releasing latent heat while they solidify.

The third type of PCM is the Eutectic compounds. These materials are expensive and toxic acids. Normally there is not any contact between the water and the PCM, but in case of failure the PCM could mix with the water, so we must not apply toxic materials. The last group is organic paraffin. These materials are suitable by the physical and chemical properties. The paraffin is obtainable at a low price. These materials have only one disadvantageous property i.e. flammability. But it is still accepted because of the presence of the water around the tube containing paraffin.

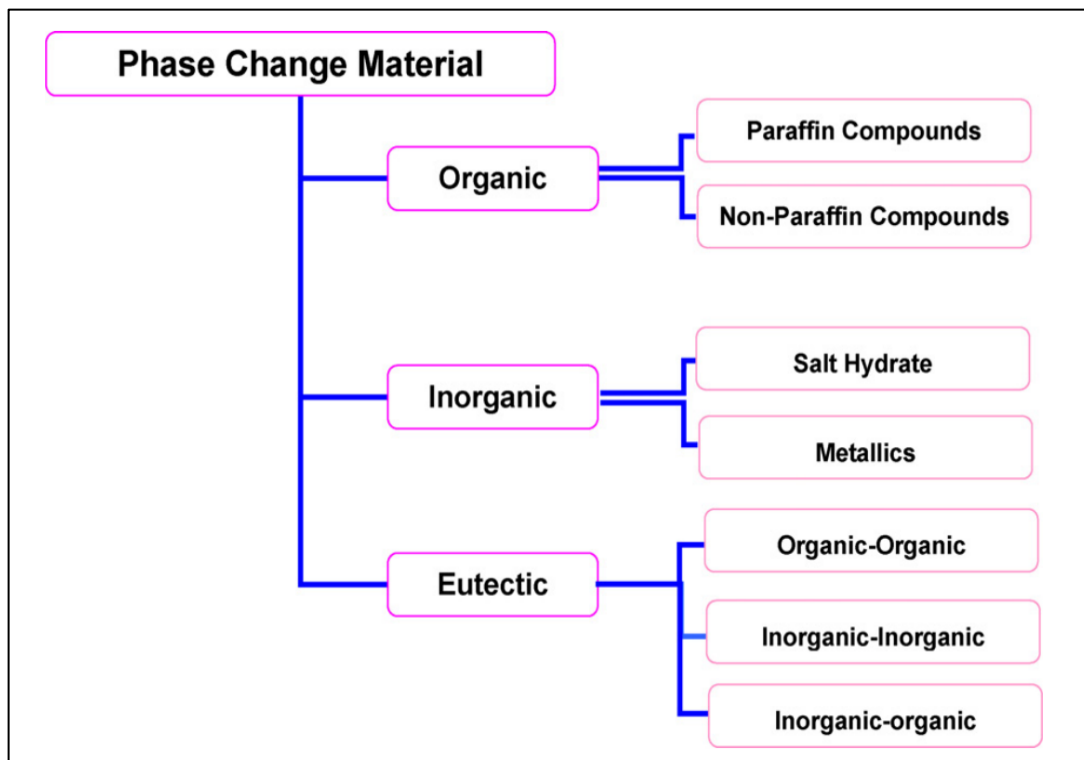


Fig 3.2. Classification of PCM

3.2.1 Desirable Properties of PCM

Different Properties from the perspective of economy, environment, chemical stability, etc. are to be considered while selecting the most suitable PCM. Figure 3.3 shows some of the properties to be taken into consideration while selecting PCM material.

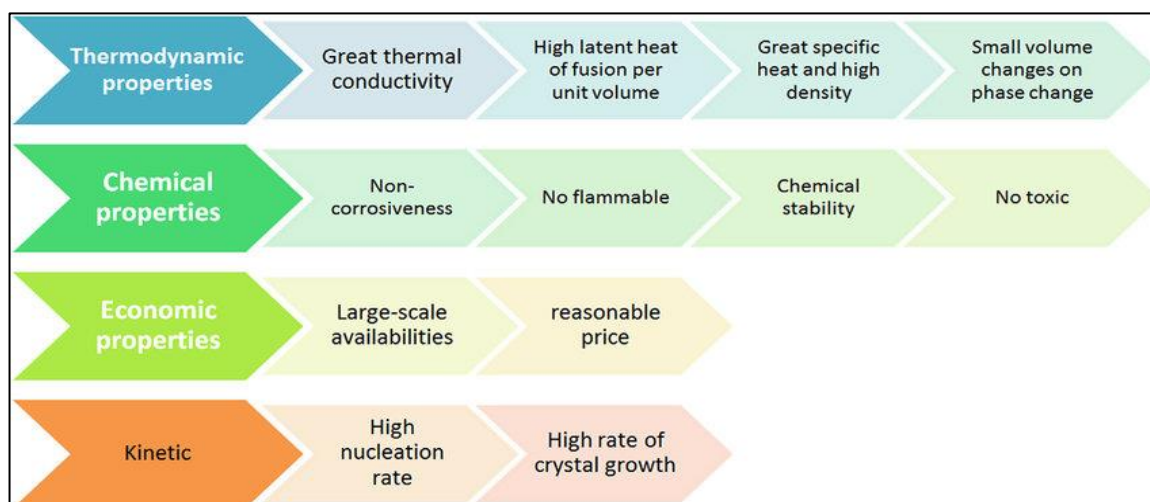


Fig 3.3. Desirable properties of PCM

The detailed study of the various properties desirable for selection of PCM, shows that the Eutectic PCMs though come with a sharp melting temperature, are the newest discovered materials and hence a large amount of test data for their various thermophysical properties is not available currently. Hence, they don't prove suitable for the current project work. Therefore, either of the Organic or Inorganic PCMs were to be used for the study and precise comparison between them regarding various parameters was studied. Table 3.1 given below shows some of the properties compared between organic and inorganic PCMs.

Table 3.1. Comparison of organic and inorganic PCM

	Organic PCM	Inorganic PCM
<u>Advantages</u>	• Chemical and thermal stability • Suffer little or no super cooling • Non-corrosives • Non-toxic • The high heat of fusion • Low vapour pressure	• High heat of fusion • Good thermal conductivity • Cheap • Non-flammables

<u>Disadvantages</u>	• Low thermal conductivity • High changes in volumes on phase change • Flammability • Lower phase change enthalpy	• Phase decomposition and suffer from loss of hydrate • Lack of thermal stability • Supercooling • Corrosion
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The comparison between organic and Inorganic PCMs concluded that Organic PCMs prove better than the inorganic ones because of their chemical and thermal stability as well as non-toxic and non-corrosive nature. On the other hand, even though Inorganic PCMs have very good thermal conductivity, they lack thermal and chemical stability and also corrode the material in contact. Hence Organic PCMs proved suitable to use in the present study.

Different types of organic PCMs were studied and compared based on the properties desired for the setup. Based on their availability and operating temperature different applications may require different PCM materials. Table 3.2 shows the comparison between the most used organic PCM materials which are studied from various literature papers.

Table 3.2. Selection of PCM

PCM Material	Operating Range for HTF (Water) (°C)	Availability	Cost	Latent Heat KJ/Kg	Specific Heat J/Kg	Thermal Conductivity (W/m.K)	Max operating temp(°C)
Paraffin Wax (OM50)	45 – 55	Easily Available	Cheap	181	2020	0.18	120
Hydrated Salt HS34	45 – 55	Not Easily Available	Costly	166	2300	0.47	80
Coconut Oil	45 – 55	Easily Available	Cheap	103	3277	0.154	54
Form Stable 42	45 – 55	Not Easily Available	Costly	181	3000	0.358	120

The comparative study between different PCM materials shows that Paraffin wax OM50 has the most suited temperature range for the present experiment as the operating temperature range for the heat transfer fluid taken in the current study is around 45 °C -55°C. Because of its easy availability, cheap price as well as the chemical stability, OM50 proves to be the most appropriate and hence selected for this project study.

3.3 Nanomaterial selection

A Nanomaterial, for being suitable for using along with PCM is required to be chemically stable, safe to use, cost effective and easily available. The most suitable Nanomaterial will make a perfect emulsion with the selected PCM and also cause ease of working to the experimenter. Table 3.3 shows a comparison between four most used Nanomaterials based on the factors like chemical reactivity, safety, Cost and Availability.

Table 3.3. Selection of Nanomaterial

Nano Particle Material	Chemical Reactivity	Safety	Cost	Availability
Graphene	Chemically stable	Material is safe	Cheap	Readily Available
Graphene Oxide	Chemically Stable	Material is safe	Expensive	Readily Available
MWCNT	Chemically Stable	Material is safe	Expensive	Not Readily available
Copper	Material reacts with PCM	Material is safe	Expensive	Readily available
Aluminium Oxide	Chemically unstable	Highly Explosive	Cheap	Readily available

The comparison between different properties of Nano particles shown in Table 3.3 shows that Graphene, Graphene Oxide and Copper are appropriate to be used along with PCM in our setup. Aluminium Oxide being highly explosive and rarely available is referred from the selection. A detailed study of the rest three materials is performed.

3.4 NEPCM Synthesis

Preparation of nanofluid is the most critical step in the experimental studies of nanofluid. There are two primary methods to prepare nanofluid: the single-step preparation process and the two-step preparation process. The single-step preparation process indicates the synthesis of nanofluids in one-step. Some of the methods in single-step preparation are Direct Evaporation method, Vacuum Evaporation onto a Running Oil method, Chemical deposition method, etc. However, in these methods, the complexity is very high.

The two-step preparation is the one where the base fluid is mixed with the Nanomaterial to get the emulsion of NEPCM. Ultrasonication is one of the methods of two-step preparation of NEPCM. In this type, the process is simpler than the single-step method. A detailed comparison between both these methods is performed in Table 3.4.

Table 3.4. Methods of synthesis of NEPCM

Process	Stability of Resulting Fluid	Complexity	Production	Quantity Produced at a time	Cost
Single-Step Process	High	Complex chemical reactions are involved	Cannot be produced in bulk quantities	20 – 50 ml	Expensive process
Two-Step Process	Moderate	Simple Process	Bulk preparation	2- 5 litres	Cheap process

The comparison between single step and two-step method proves that the two-step method is the most suitable method for the current project setup. Because of its simplicity, cheaper cost, and ability to produce bulk quantities Two-step method is selected for the synthesis

of NEPCM. In the two-step preparation method, Ultrasonication is the most used method according to the studies and market research. It is the most easily available and cheapest method for the synthesis of NEPCM for experimental purposes.

The preparation of the NanoPCM process by the two step process have been illustrated in the figure 3.4 . The sample is measured according to the requirement and then poured into the beaker. Here the first step is the mixing of the nanofluid and for this the sample underwent magnetic stirring. A magnet kept inside the beaker rotates at high rpm that helps stir the fluid.

After 1 hour of magnetic stirring, the sample has been transferred to the ultrasonication bath which is the 2nd step of the preparation. The sonicator aids in deagglomeration of the nanoparticle and it evenly disperses graphene by breaking the aggregates of colloidal suspension. During the entire process the sample have been maintained at a temperature higher than its melting point.

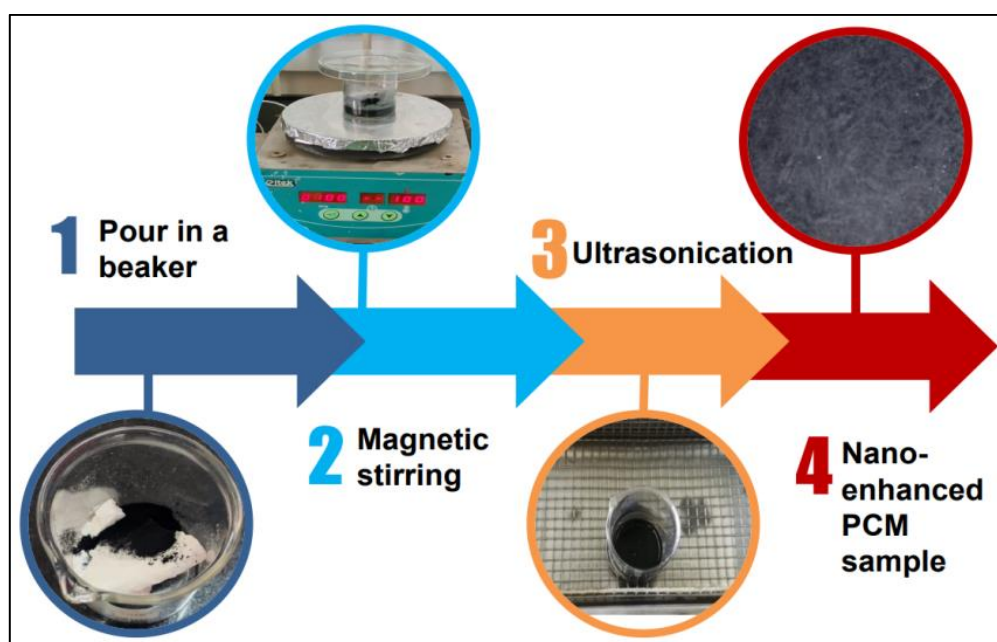


Fig 3.4. Two step process preparation

Sonication uses an ultrasonic bath or probe to apply sound energy to a liquid containing particles. Ultrasonic frequencies (>20 kHz) are usually used, so the process is also known as Ultrasonication. Apparatus applied in laboratories can be used for volumes up to 2 litres, whereas industrial Sonicators are used in the process of development of up to 2000 litres. Figure 3.4. shows the apparatus of Ultrasonication along with the probe attached to it.



Fig 3.5. Ultrasonication Apparatus

Synthesis of Nano enhanced PCM will be performed by taking 3-5% by volume nano particles. Ultrasonication of Graphene nano particles and PCM (OM-50) will be carried out using a water bath Sonication apparatus. The apparatus will be run for about 45 minutes for each batches of NePCM while keeping the temperature of water bath above 50°C.

3.5 NEPCM Stability

Stabilization of nanofluids is the most urgent concern. It means the nanoparticles do not agglomerate at a significant speed. The reason for the agglomeration of the nanoparticle is that nanoparticles in nanofluid are able to attach together as a result of the attractive force against the repulsive force. Respectively, they are van der Waals and electrical double layer, which exist between nanoparticles. Also, they come close to others by reason of the random Brownian motion. When attraction prevails over repulsion, two nanoparticles will collide, and nanoparticles tend to agglomerate in clusters.

Stabilization of nanofluid can be estimated through measurement of the exact amount of the settled and suspended nanoparticles in BF over time. There are six different methods to study the stabilization of nanofluids as follows: UV–Vis spectrophotometry, Zeta potential test, Sedimentation photograph capturing, Scanning electron microscopy and Transmission electron microscopy, sedimentation balance method etc.

3.5.1 Zeta Potential Test

Zeta potential is a measure of the effective electric charge on the nanoparticle's surface, quantifying the charges. When a nanoparticle has a net surface charge, the charge is screened by the concentration of ions of opposite charge near the nanoparticle surface. This layer of oppositely charged ions moves with the nanoparticle and together with the layer. The zeta potential is a measure of the difference in potential between the bulk fluid in which a particle is dispersed and the layer of fluid containing the oppositely charged ions that is associated with the nanoparticle surface. Particles with a positive zeta potential will bind to negatively charged surface and vice versa.

The magnitude of the zeta potential provides information about particle stability. The higher the magnitude of potential exhibits increased electrostatic repulsion and therefore increased stability. It is important to consider that the magnitude of the charge on the nanoparticle surface depends on the solution pH. Zeta potential is measured by adding a solution to a cell that contains two gold electrodes. When a voltage is applied to the electrode, the particles will move toward the electrode with the opposite charge. A Doppler technique is used to measure the particle velocity as a function of voltage. A laser passes through the cell, and as particles move through the laser beam, the intensity of scattered light fluctuates at a frequency proportional to the particle speed. The particle speed at multiple voltages is measured, and this data is used to calculate the zeta potential.

In nanofluid, zeta (ζ)-potential is the difference of potential between the immobile sheet of BF grafted to the nanoparticle and BF. The electrical potential is considered as ζ -potential that is associated with the repulsive force between neighboring, analogously charged nanoparticles in the suspension system.

Nanofluids are believed to have good stabilization with ζ -potential between 40 and 60 mV. Nanofluids with ζ -potentials higher than 60 mV are known as having excellent stabilization. The stability behaviour of the nanofluid to its corresponding value of zeta potential is outlined in the table.

Table 3.5. Zeta Potential Stability Behavior

Zeta potential (mV)	Stability behavior
0 to ± 5	Rapid coagulation or flocculation
± 10 to ± 30	Incipient instability
± 30 to ± 40	Moderate stability
± 40 to ± 60	Good stability
> 61	Excellent stability

3.6 Analytical Calculation

A precise analysis of the system is required to validate the theoretical and practical conclusions with one another and also to work on relative aspects. For example, Reynold's number has been calculated for selection of the inner pipe diameter. Hence this section depicts the analytical calculations were performed to study some parameters in order to compare them with practical values.

3.6.1 Reynolds Number

The heat transfer coefficient increases with an increasing Reynolds number, which is a function of density, flow velocity, hydraulic diameter, and viscous coefficient. Moreover, the convective heat transfer coefficient is a function of the Nusselt number, the conductive heat transfer coefficient, and the hydraulic diameter. In order to increase the convective heat transfer coefficient, it is necessary to increase the Reynolds number by increasing flow velocity and hydraulic diameter, but increasing hydraulic diameter also decreases the convective heat transfer coefficient. Therefore, the most important design goal is to increase the flow velocity, with smallest diameter which strictly improves heat transfer. Table 3.6 shows the calculations of flow velocity and Reynold's Number for different diameters of copper pipe. Formula for finding Reynolds Number is given by Eq.3.2.

- $$\text{Reynolds Number} = \frac{\text{Fluid density} * \text{Velocity} * \text{Diameter}}{\text{Dynamic viscosity}}$$

SAMPLE CALCULATION-

1) Inner Pipe Diameter –

$$(Di_1) - 0.5 \text{ inch} = 0.5 \times 25.4 = 12.7 \text{ mm} = 0.0127 \text{ m}$$

Thickness – 3.3 mm

$$(Do_1) - 12.7 + 3.3 = 16 \text{ mm} = 0.016 \text{ m}$$

- Reynolds Number = $\frac{\text{Density} \times \text{Velocity} \times \text{Diameter}}{\text{Dynamic viscosity}}$
$$= \frac{1000 \times 0.2605 \times 0.0127}{0.00054}$$
$$= 6126.57$$

Table 3.6. Variation of Reynolds number with diameter of inner pipe

Volume Flow Rate (LPH)	Volume Flow Rate (m ³ /s)	Diameter of inner pipe (Inch)	Diameter of inner pipe (m)	Area of Cross-Section (m ²)	Velocity of Water (m/s)	The viscosity of Water (kg/ms)	Reynolds Number
120	0.000033	0.5	0.0127	0.00013	0.2605	0.00054	6126.57
120	0.000033	0.75	0.019	0.00029	0.1163	0.00054	4092.03
120	0.000033	1	0.0254	0.00051	0.0651	0.00054	3062.11
120	0.000033	1.5	0.0381	0.00114	0.0289	0.00054	2039.05
120	0.000033	2	0.0508	0.00203	0.0164	0.00054	1533.70

Pipe of diameter 0.5 inches can be selected based on table, as it has maximum Reynolds number for a given flow rate.

- 2) Intermediate Pipe Diameter –

$$(Di_2) - 2 \text{ Inch} = 2 \times 25.4 = 50.8 \text{ mm} = 0.0508 \text{ m}$$

Thickness – 3.3 mm

$$(Do_2) - 50.8 + 3.3 = 54.1 \text{ mm} = 0.0541 \text{ m}$$

Based on optimum area of cross section of intermediate pipe, in order to store PCM and NePCM, we have considered the diameter as 2 inch.

$$\begin{aligned}\text{Reynolds Number} &= \frac{\text{Density} * \text{Velocity} * \text{Hydraulic Diameter}}{\text{Dynamic viscosity}} \\ &= \frac{1000 * 0.0538 * (0.0762 - 0.0541)}{0.00054} \\ &= 2414.74\end{aligned}$$

Table 3.7. Variation of Reynolds number with diameter of outer pipe

Volume Flow Rate (LPH)	Volume Flow Rate (m ³ /s)	The Diameter Of pipe (Inch)	Diameter of pipe (m)	Hydraulic diameter (m)	Area of Cross-section (m ²)	Velocity of Water (m/s)	The Viscosity of Water (kg/ms)	Reynolds Number
480	0.00013	3	0.0762	0.0221	0.0022	0.0538	0.00054	2414.74
480	0.00013	3.5	0.0889	0.0348	0.0039	0.0341	0.00054	2203.90
480	0.00013	4	0.1016	0.0475	0.0058	0.0229	0.00054	2022.90
480	0.00013	4.5	0.1143	0.0602	0.0079	0.0167	0.00054	1868.11
480	0.00013	5	0.127	0.0729	0.0103	0.0129	0.00054	1749.97

Pipe of diameter 3 inch can be selected based on table above as it has maximum Reynolds number for a given flow rate.

Chapter 4

Simulation

Design of heat exchanger is crucial to check how effectively heat transfer takes place between PCM and hot water. In order to ascertain the experimental readings performed using physical model, we need to simulate the results too. Hence Thermal Analysis and simulation of NEPCM heat exchanger is done using ANSYS fluent in workbench.

4.1 Input Parameters

Different thermal properties of PCM as well as Nano particles were referred and liquefaction of PCM as well as NEPCM were simulated. The results of both the systems were compared and the liquefaction time taken by them were analyzed.

Table 4.1. shows the thermal properties of PCM OM-50 which were referred from the PCM datasheet. The same values were used as input parameters during the simulation of PCM as well as NePCM systems in ANSYS Fluent.

Table 4.1. Simulation Data for PCM (OM- 50)

Sr. no.	Material properties	Value of Quantity	Unit
1	Density	961	kg/m ³
2	Specific Heat	3330	J/kg-K
3	Thermal Conductivity	0.18	W/m-K
4	Viscosity	0.00181	kg/m-s
5	Thermal Expansion Coefficient	0.00012	1/K
6	Pure Solvent Melting Heat (Latent heat)	170000	J/kg
7	Solidous Temperature	323	K
8	Liquidous Temperature	326	K

After addition of Graphene Nanoparticles in the regular PCM, the resulting NEPCM mixture is formed with transformed thermal properties. The revised thermal properties of NEPCM were calculated using the following set of formulae:

$$1. \text{ Density: } \rho_{\text{npcm}} = \Phi \rho_{\text{np}} + (1 - \Phi) \rho_{\text{pcm}}$$

$$2. \text{ Specific Heat : } C_{p_{\text{npcm}}} = (1 - \Phi) C_{p_{\text{pcm}}} + \Phi C_{p_g}$$

$$3. \text{ Thermal Conductivity : } K_{\text{npcm}} = K_{\text{pcm}} \frac{[K_g + 2 K_{\text{pcm}} - 2 \Phi(K_{\text{pcm}} - K_g)]}{[K_g + 2K_{\text{pcm}} + \Phi(K_{\text{pcm}} - K_g)]}$$

$$4. \text{ Viscosity : } \mu_{\text{npcm}} = \frac{\mu_{\text{pcm}}}{(1 - \Phi)^{2.5}}$$

$$5. \text{ Latent Heat : } L_{\text{npcm}} = \frac{(1 - \Phi) * \rho_{\text{pcm}} * L_{\text{pcm}}}{\rho_{\text{npcm}}}$$

Table 4.2. shows the calculated values of thermal properties of NePCM which were acquired by using the above mentioned formulae. The same values were used as input parameters for the simulation study of NePCM system in ANSYS Fluent.

Table 4.2. Simulation Data for NePCM

Sr. no.	Material properties	Value of Quantity	Unit
1	Density	1088	kg/m ³
2	Specific Heat	3163.53	J/kg-K
3	Thermal Conductivity	0.21	W/m-K
4	Viscosity	0.00205	kg/m-s
5	Thermal Expansion Coefficient	0.00012	1/K
6	Pure Solvent Melting Heat (Latent heat)	142654	J/kg
7	Solidous Temperature	323	K
8	Liquidous Temperature	326	K

4.2 Geometry

The Cad model of the Heat exchanger is shown in the figure 4.1. It consists of three concentric pipes connected by end caps at the end. The dimensions of the heat exchanger were taken according to the design calculations.

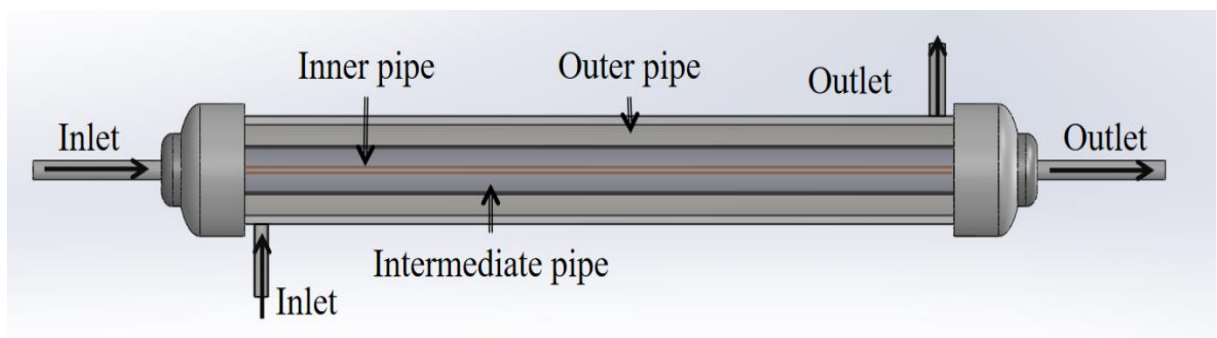


Fig 4.1. Lateral transparent section

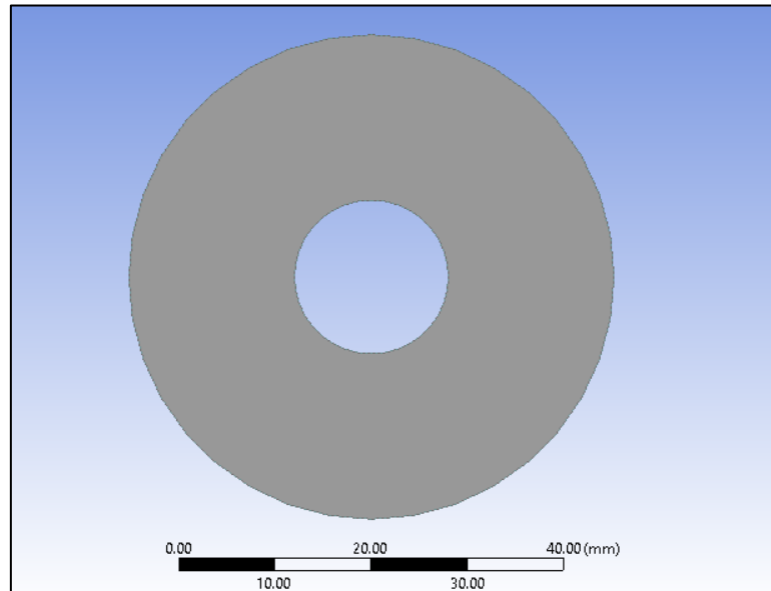


Fig 4.2. 2D Geometry for simulation

4.3 Meshing

The geometry was then proceeded for meshing. Various meshing properties were set with a view of maximizing the accuracy as well as minimizing the time for simulation as much as possible. The size function was kept curvature in order to avoid the anomalies and the order of the element was kept linear. The meshing was kept fine for acquiring precise results while the smoothing properties were kept high for getting a clear meshing. The element type was kept 4 node quadrilaterals in order to keep the errors as less as possible. The number of nodes were set to 2100 and the number of elements were 2000. Table 4.3. shows the simulation settings done for Meshing in ANSYS.

Table 4.3. Simulation setting for Meshing

Display	
Display Style	Using geometry settings
Defaults	
Physics Preference	CFD
Solver Preference	Fluent
Export Format	Standard
Export Preview Surface Mesh	No
Element Order	Linear
Sizing	
Size Function	Curvature
Max Face Size	Default (3.59210 mm)
Growth Rate	Default (1.20)
Min Size	Default (3.5921e-002 mm)
Quality	
Check Mesh Quality	Yes, Errors
Target Skewness	0.9
Smoothing	High
Min	3.0877e-002
Max	0.45262
Average	0.18571
Standard Deviation	9.5182e-002

The mentioned parameters were used as input for meshing and the meshing was progressed. Figure 4.3. shows the image of the geometry after the meshing was performed. The meshed model was used along with the surrounding parameters and the further results were acquired.

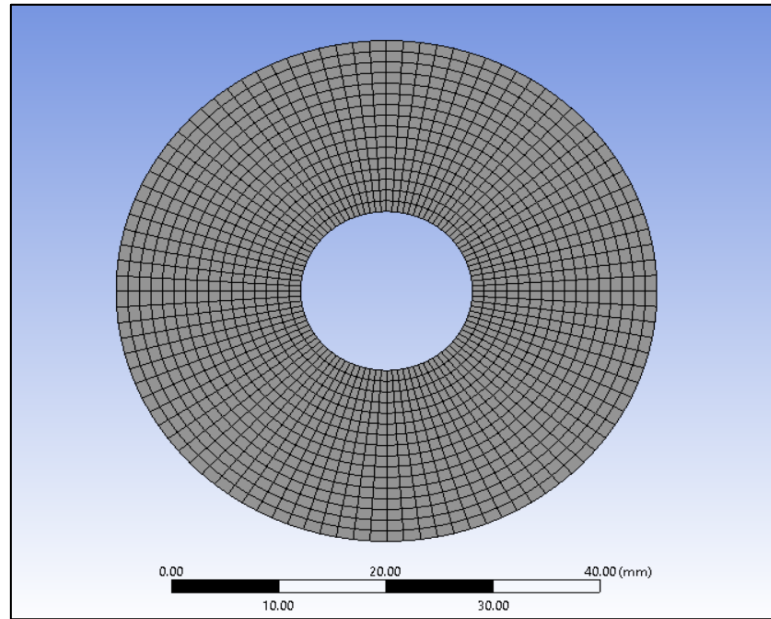


Fig 4.3. Meshing

4.3 Simulation Results

Using the input parameters for PCM, the simulation was progressed. The results were acquired and the liquefaction of PCM system was analyzed. The simulation results were carried out for different amount of nanomaterials.

The melting and solidification phenomenon of the PCM in annular space between the concentric pipes have been shown in the figure below. Figure 4.4 and figure 4.5 shows the melting phenomenon and solidification phenomenon respectively at various time interval. The amount of liquid fraction indicates the PCM melted in the annular region. At time equal to 600 seconds, the liquid fraction was found out to be 0.08 while at time 1950 seconds, 4700 seconds and 7200 seconds, the liquid fraction was found out be 0.25, 0.60 and 0.92 respectively.

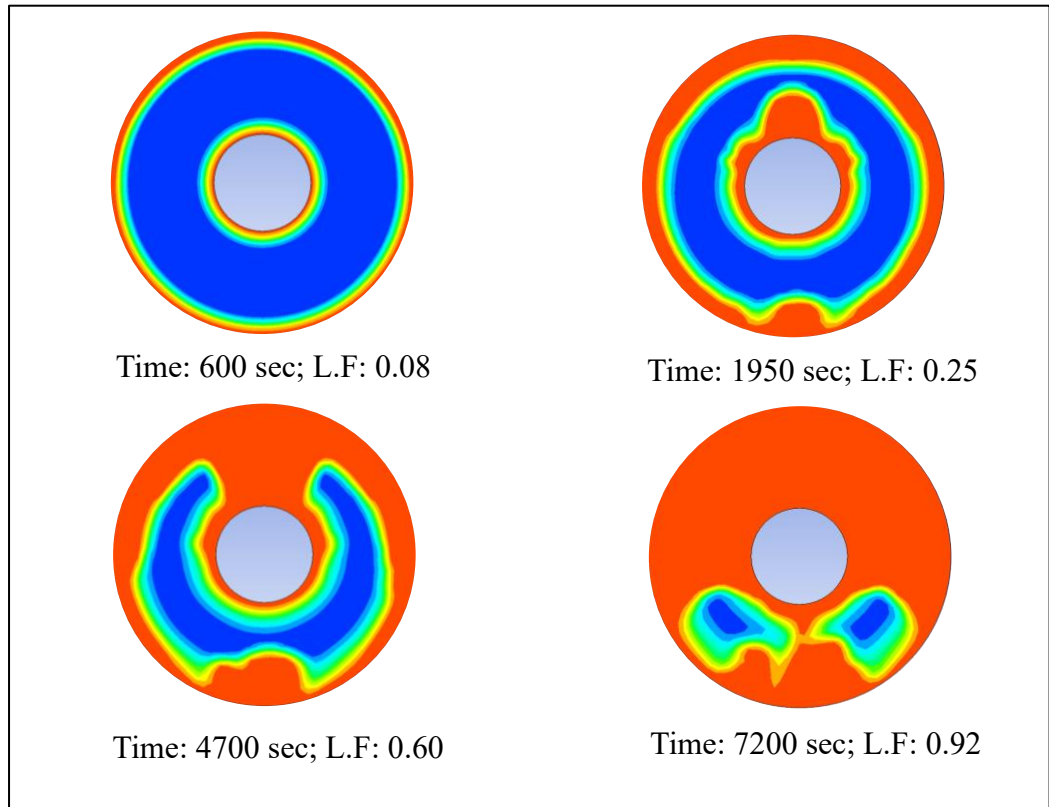


Fig 4.4. Melting phenomenon

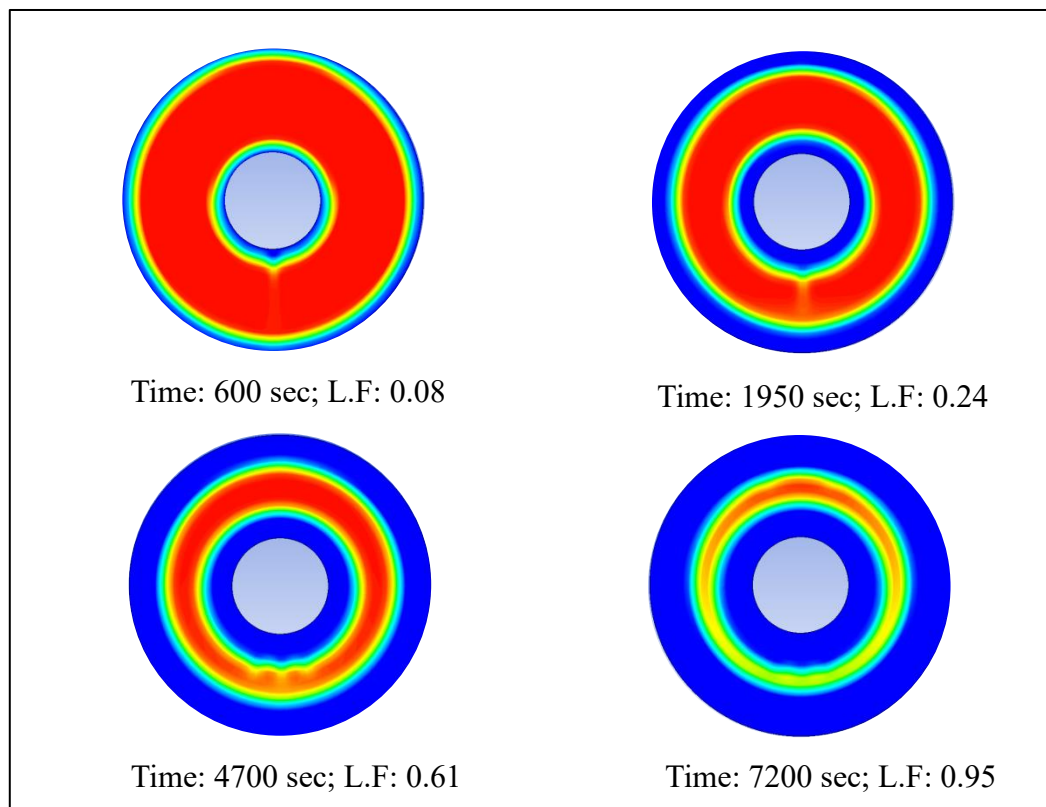


Fig 4.5. Solidification phenomenon

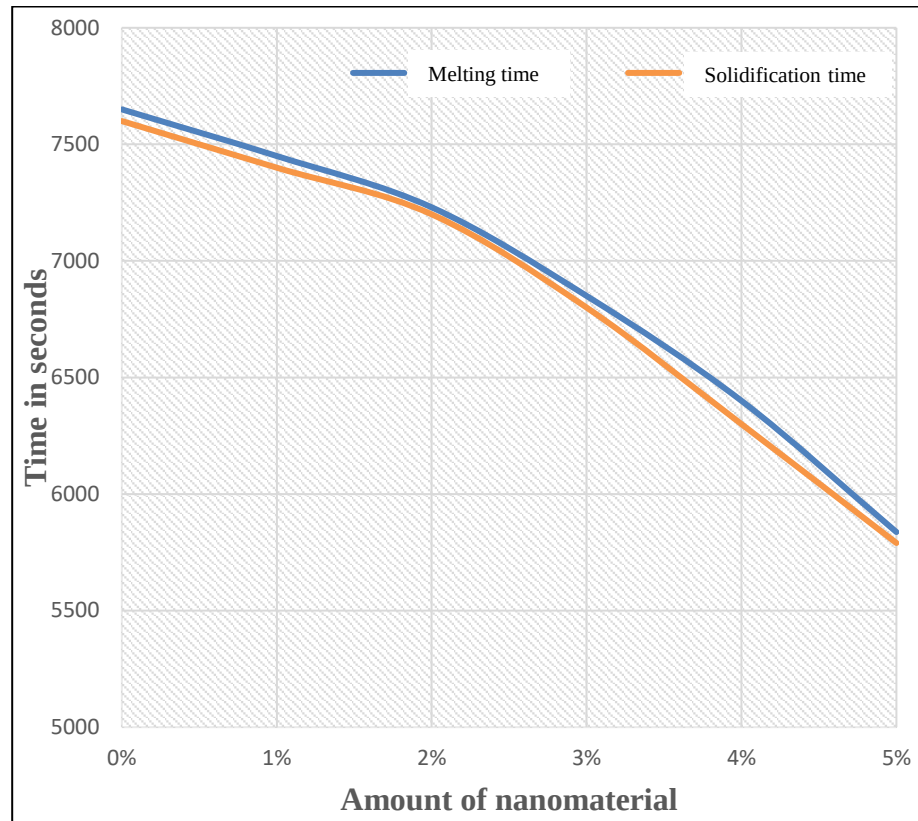


Fig 4.6. Time required for melting and solidification

The simulation was carried out for different amount of nanomaterials and the resulting time of resulting emulsion of base PCM and nanomaterial. It was found that as we increase the amount of nanomaterial the time required for melting and solidification decreases. The least time was found at 5% amount of nanomaterial which was around 5800 seconds and the highest was found in normal PCM which was around 7600 seconds.

Chapter 5

Experimental Setup

5.1 CAD Model

CAD Modelling of any system is the initial stage of the design procedure. Before the actual experimental setup is fabricated, a computer aided designed model is prepared with a view of validating and analyzing the errors or limitations that practical model might face.

The Computer aided model for the proposed experimental setup was designed as shown in Figure 5.1. It was made after considering the dimensions calculated in the previous sections. The following figure consists of the CAD model of the proposed design of the experimental setup modelled by using Autodesk Inventor.

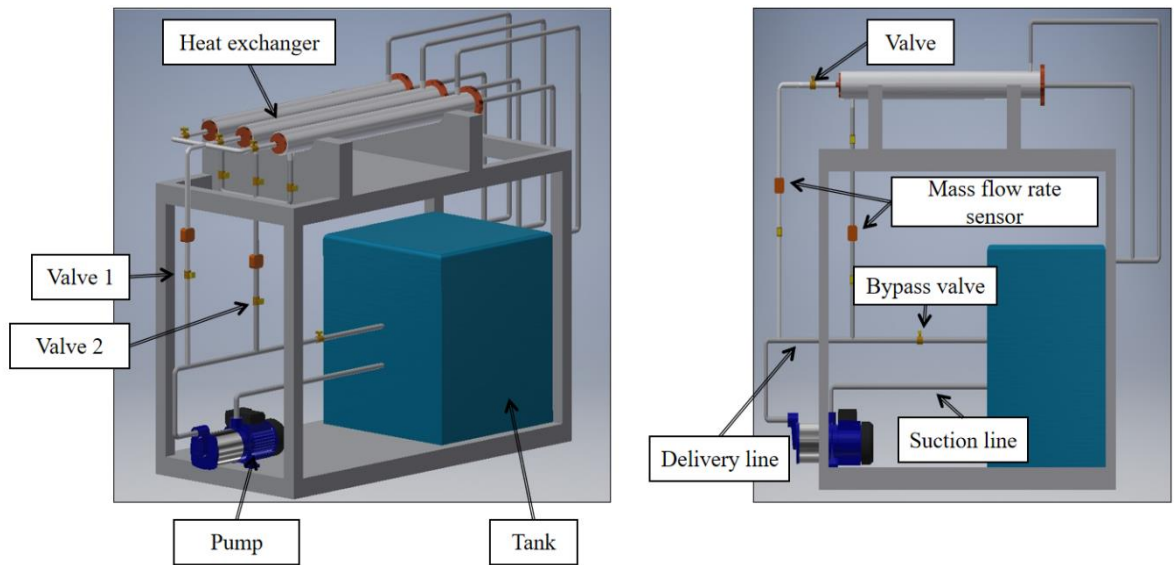


Fig 5.1. CAD model of apparatus

The proposed CAD model consists of a triple tube type of heat exchanger unit with an inlet and outlet conditions for heat transfer fluid i.e. water through the inner and outer copper pipe. The space between the two pipes will consist of NEPCM material. Water heated with the help of an immersion coil will be pumped to the heat exchanger unit. Electromagnetic Flow sensor is installed to note the flow rate which can be controlled by using valves at different locations. Thermocouples are installed at various locations to measure the respective temperatures. A sump tank is included in order to store water and acquire a uniform flow.

Experimentation of the Normal PCM based heat exchanger was performed for the charging cycle. Charging of the PCM is the process where the PCM acquires heat from the HTF and melts to liquid form. The water in the tank was heated to the predetermined temperature using the heating coils. The temperature of the water was maintained throughout the process with the help of the thermostat. The hot water is pumped through the inner and outer pipe of the system. It is continued until all the thermocouples show the temperature slightly higher than the melting point of PCM. Once this stage is reached, it indicates that the PCM inside the heat exchanger has completely changed its phase from solid to liquid.

Various parameters were fixed during the charging of the system. Following table shows the parameters that were considered in the study for the charging of the system.

Table 5.1 Experimental parameters

Experimental parameters	Inner Pipe	Outer pipe
Water Flow rate (LPH)	120	480
Hot water inlet temperature (°C)	60	60

Temperatures were measured using K-type thermocouples which were inserted at the predetermined locations on the pipe system. The readings were obtained using the NI DAQ system to which all the thermocouples were connected. Figure 5.2 shows the placement of the thermocouples with reference to the heat exchanger.

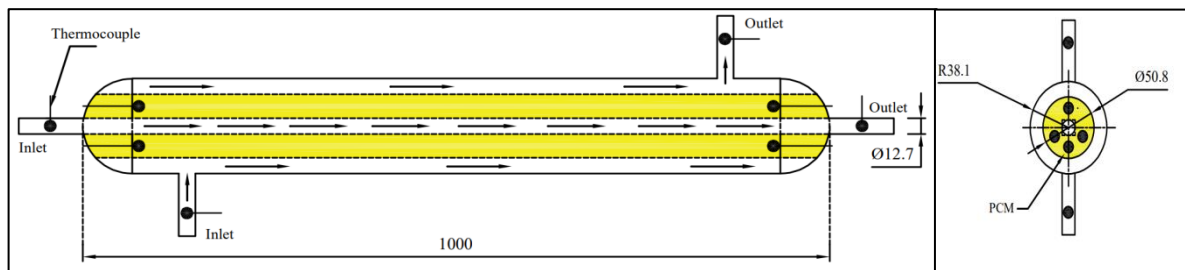


Fig 5.2. Location and arrangement of thermocouples

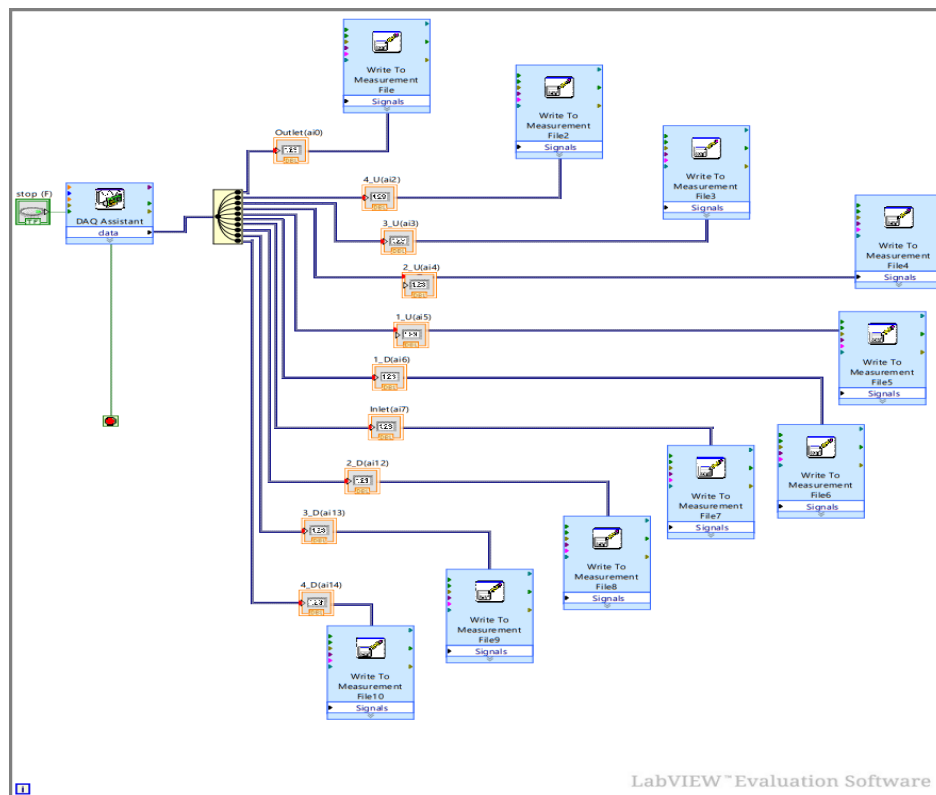


Fig 5.3. Thermocouple block diagram

5.2 Schematic Diagram

A schematic diagram was created to represent the elements of a system using abstract, graphic symbols rather than realistic pictures. It usually omits all details that are not relevant to the key information and is intended to convey the same meaning as the actual system, and includes oversimplified elements in order to make the essential meaning easier to grasp. Figure 5.4. shows the schematic diagram of the storage system in the current project.

The same flow system was considered for fabricating the actual setup. Based on the study of CAD model and the schematic diagram the procurement of components and their fabrication was performed.

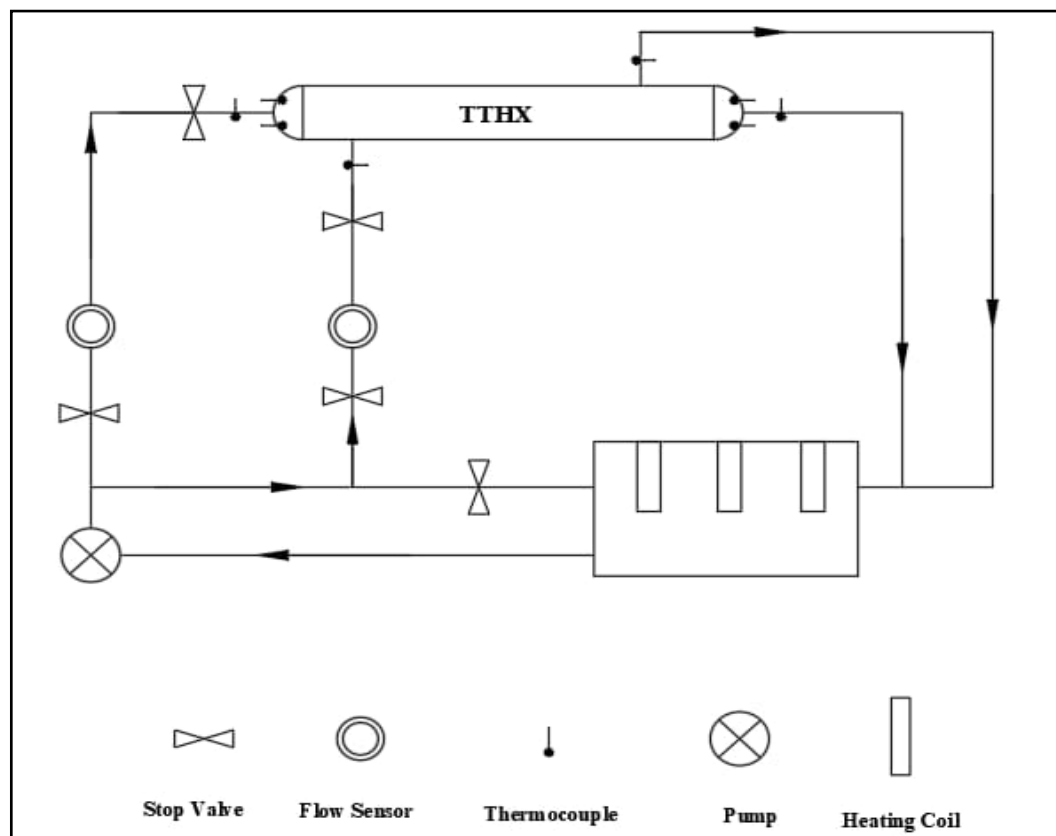


Fig 5.4. Schematic Diagram

5.3 Selection of Components

1. Immersion Heating Coil: Immersion rods or Immersion water heaters are one of the most convenient way of heating water. It is a simple appliance which can heat a bucket full of water in 15-20 minutes. It comes with a heating element made of copper in a coiled shape. This will be used for heating the water in the sump tank before pumping throughout the setup. Figure 5.5. shows the Immersion heater appliance used at domestic purposes.

Specifications: Power consumption: 2000W

Heater material: Copper

Operating Voltage: 220-240volts.



Fig 5.5. Immersion Water Heater

2. UPVC pipe: UPVC is a Thermoplastic material derived from common salt and fossil fuels. UPVC has high chemical resistance across its operating temperature range, with a broad range of operating pressures. It can also be suitable as an insulating material for minimizing drop of temperature of water. Figure 5.6. shows the UPVC pipes used in domestic applications.

Specifications: Pipe size: ½ and 3 inches.

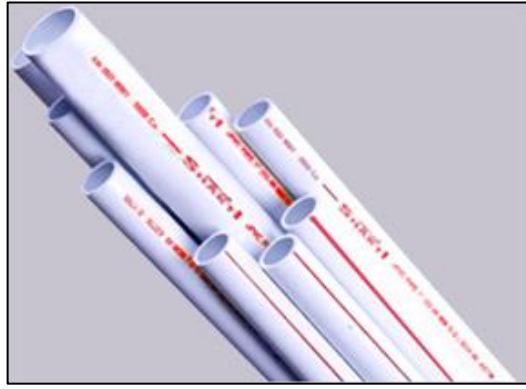


Fig 5.6. UPVC Pipes

3. UPVC Ball Valve: Flow control is done by using valves. A ball valve is a form of quarter-turn valve which uses a hollow, perforated and pivoting ball to control flow through it. It is open when the ball's hole is in line with the flow and closed when it is pivoted 90-degrees by the valve handle. It is very easy and convenient to use and reliable in operation. Figure 5.7. shows the actual image of a UPVC Ball valve.

Specifications: Valve size: ½ inches



Fig 5.7. UPVC Ball Valve.

4. Copper pipe: Copper has many desirable properties for thermally efficient and durable heat exchangers. First and foremost, copper is an excellent conductor of heat. This means that copper's high thermal conductivity allows heat to pass through it quickly. Other desirable properties of copper in heat exchangers include its corrosion resistance, maximum allowable stress and internal pressure, hardness, thermal expansion, specific heat, high melting point, alloy ability, ease of fabrication, and ease of joining. Figure 5.8 shows copper pipes of standard size used in many heating applications.

Specifications: Pipe size: ½ and 2 inches.



Fig 5.8. Copper Pipes.

5. Thermocouple: A thermocouple is an electrical device consisting of two dissimilar electrical conductors forming an electrical junction. A thermocouple produces a temperature-dependent voltage as a result of the thermoelectric effect, and this voltage can be interpreted to measure temperature. Thermocouples are a widely used type of temperature sensor. Among various types of thermocouple, type-K was used for the experiment. Type-K (chromel–alumel) is the most common general-purpose thermocouple with a sensitivity of approximately $41 \mu\text{V}/^\circ\text{C}$. It is inexpensive, and a wide variety of probes are available in its -200°C to $+1350^\circ\text{C}$ range. Figure 5.9. shows a typical type-K thermocouple.

Specifications: Type-K

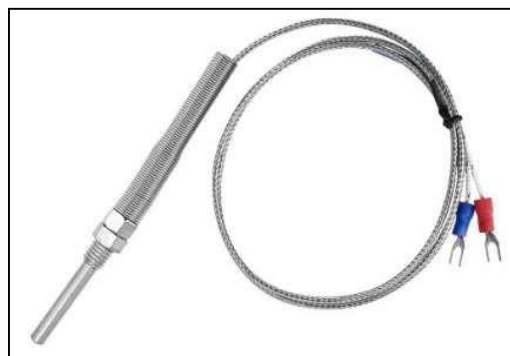


Fig 5.9. Type-K Thermocouple.

6. Water Pump: A pump is a device that moves fluids (liquids or gases), by mechanical action. It is most commonly used in residential as well as commercial buildings to raise the water from ground level to upper levels. Capacity of pumps vary from model to model and a suitable pump can be chosen from standard available capacities. Figure 5.10. shows domestic pump.

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Specifications: Model: Lakshmi Pump

Power: 370 Watt.

Voltage: 230 volts

Current: 2 Amps Max.

Maximum capacity: 360 LPH.



Fig 5.10. Water Pump.

7. Sump tank: Storage of excess water and heating of water is done in a sump tank of 100 litres capacity. After one complete cycle, water comes back to the sump tank by travelling through the pipes and the pump. Figure 5.1. shows a HDPE sump tank used for many domestic and industrial purpose.

Specifications: Material: HDPE (High Density Polyethylene),
Capacity: 100 litres.



Fig 5.11. Sump Tank

8. Electronic Flow sensor: Flow measurement is the quantification of bulk fluid movement. Flow can be measured in a variety of ways. In present study, Hall effect sensor is used to measure flow rate of water with a working range of 30L/min. Figure 5.12. shows an actual image of YF-S201 Electronic flow sensor used in many flow measurement applications.

Specifications: Working Range : 1-30 L/min.

Water Pressure : <1.75 MPA.



Fig 5.12. Electronic Flow Sensor

9. Asbestos Insulation Tape: Asbestos tape is used for insulating the outer (GI) pipe. Asbestos material has excellent electrical and heat insulating properties and high tensile strength. Also, it is durable and economical, high in water repellence, low in water absorption, resistant to flame, heat, oil, fungus and mould, non-corrosive to cables, non-sagging and abrasive resistant. The tape is available in various sizes but 1.5 inch width was taken for our application considering the cost and availability. Figure 5.13. shows the actual image of the Asbestos insulating Tape used in many industrial applications.

Specifications: Material : Asbestos

Width : 30 mm

Thickness : 3 mm



Fig 5.13. Asbestos Insulation Tape.

10. Thermostat: A thermostat is a component which senses the temperature of a physical system and performs actions so that the system's temperature is maintained near a desired setpoint. Thermostats are used in any device or system that heats or cools to a setpoint temperature. A thermostat exerts control by switching heating or cooling devices on or off to maintain the correct temperature. Figure 5.14. shows the image of geyser thermostat which will be attached in line with the immersion coils for the current setup.

Specifications: Material : Brass.

Length:11inches.

Diameter : 10 mm

Operating Range : 0°C -75°C



Fig 5.14. Thermostat

5.4 Fabrication of Apparatus

Using calculated design values, the experimental setup is fabricated. All the mentioned components are used for the fabrication of the apparatus.

The triple tube heat exchanger consists of three pipes the inner pipe , the intermediate and the outer pipe which again consists of three different construction like finned tube, concentric tube and concentric tube with nanomaterial. So the basic fabrication consists of drilling and the boring operations of the end caps which are present at the 2 ends of the heat exchanger, 4 holes were drilled on the end cap for the thermowells to fit in it. Now, the inner pipe was placed carefully and cap was placed at one end followed by the intermediate tube. At this stage liquid PCM is poured into the intermediate pipe consisting of the inner pipe too and the other end of the pipes are covered with the end cap. Now, the outer pipe is placed over the 2 pipes by placing larger end caps at both the ends and appropriate holes are drilled on the outer ends of the pipe for inlet and outlet.

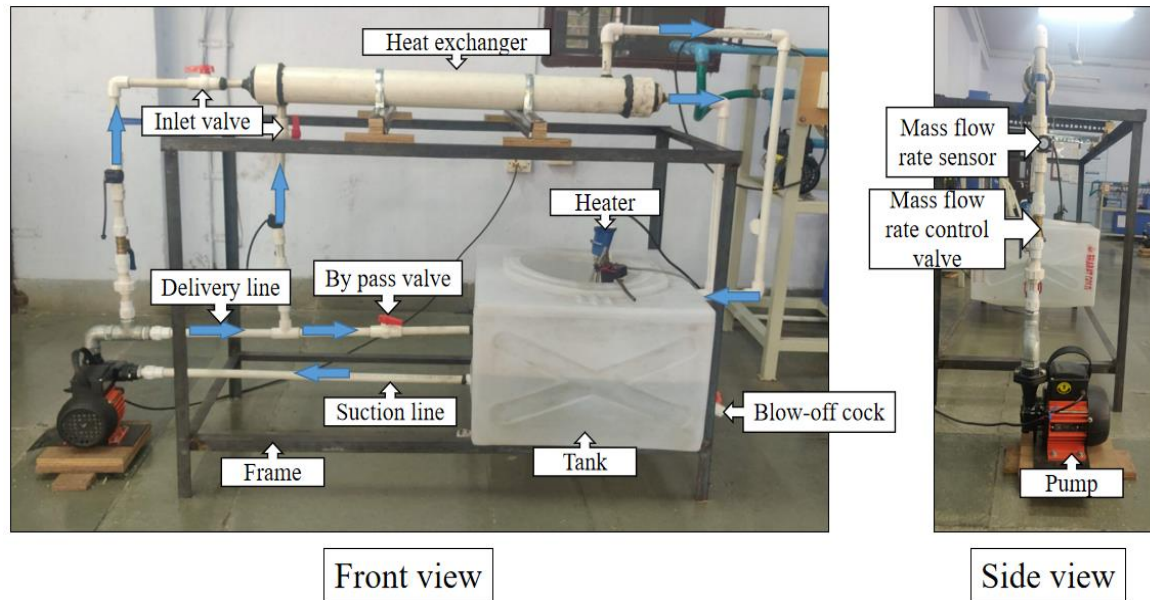


Fig 5.15. Fabricated triple tube heat exchanger

Pipes, sump tank and pump are supported on a frame manufactured using MS angles welded together in measured lengths. As shown in Fig 5.15, We have Manufactured triple tube heat exchanger with the piping and mounted on the MS angle frame. In piping, we attached the pump and tank with 2 pipes (i.e. one for flow from tank to pump and second for back flow from pump to tank). We attached two more PVC pipe to the back flow pipe (i.e second inlet of heat exchanger in outer PVC pipe) for circulating the water to the first inlet (i.e. the inner pipe) and the second inlet (i.e. the outer PVC pipe). The outlet of outer PVC pipe and inner pipe are directly connected to the tank. Pump is mounted near the sump tank to maintain the flow of water throughout the apparatus. On both the inlet pipes we have attached mass flow rate sensor for reading the flow rate through the pipes and Gate valves for adjusting flow of water. Immersion coils are attached inside the lid of the sump tank in order to heat the water.

Chapter 6

Results and discussion

6.1 Zeta Potential Test Results

The zeta potential is a key indicator of the stability of colloidal dispersions. The magnitude of the zeta potential indicates the degree of electrostatic repulsion between adjacent, similarly charged particles in a dispersion. For molecules and particles that are small enough, a high zeta potential will confer stability, i.e., the solution or dispersion will resist aggregation. When the potential is small, attractive forces may exceed this repulsion and the dispersion may break and flocculate. So, colloids with high zeta potential (negative or positive) are electrically stabilized while colloids with low zeta potentials tend to coagulate or flocculate.

Table 6.1. Zeta Potential Results

Sr no.	Sample	Temperature (in °C)	zeta potential (in mV)	Average
1	0.05	25	-31.2	-30.26666667
2	0.05	25	-30.4	
3	0.05	25	-29.2	
4	0.045	25	-44.4	-42.2
5	0.045	25	-40.8	
6	0.045	25	-41.4	
7	0.04	25	-45	-44.93333333
8	0.04	25	-46.3	
9	0.04	25	-43.5	
10	0.035	25	-48.3	-48.43333333
11	0.035	25	-49.8	
12	0.035	25	-47.2	

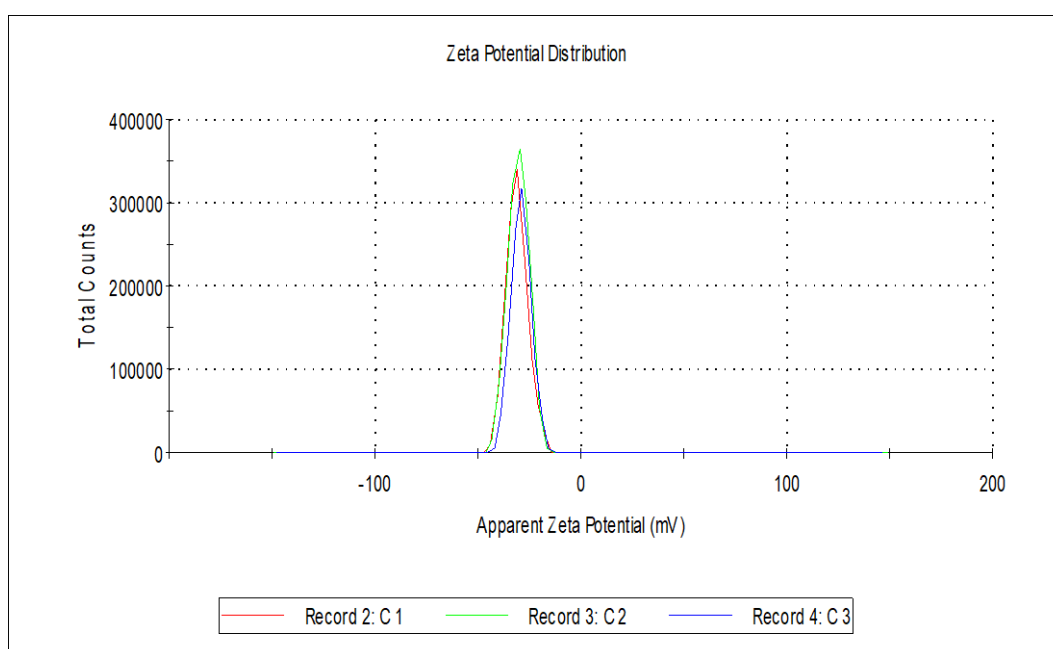


Fig 6.1. Zeta Potential Sample 5%

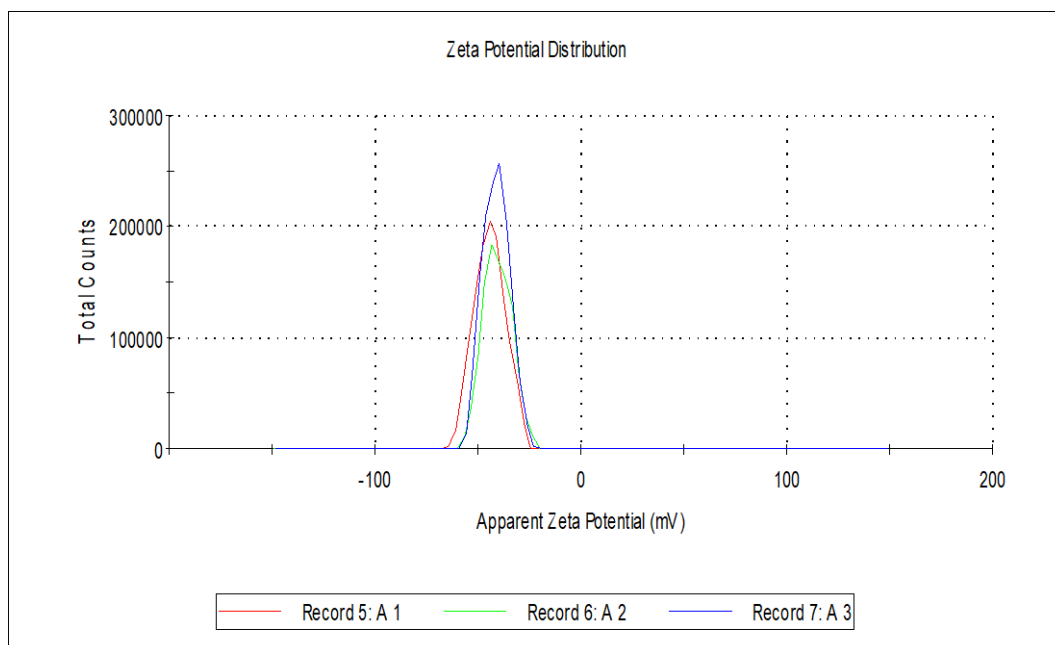


Fig 6.2. Zeta Potential Sample 4.5%

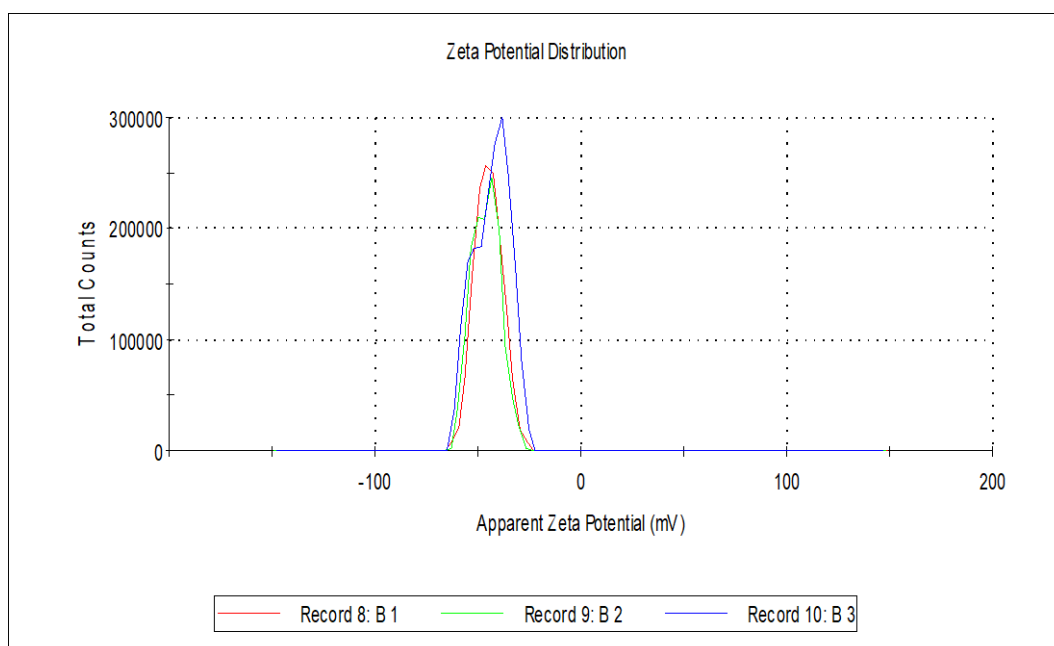


Fig 6.3. Zeta Potential Sample 4%

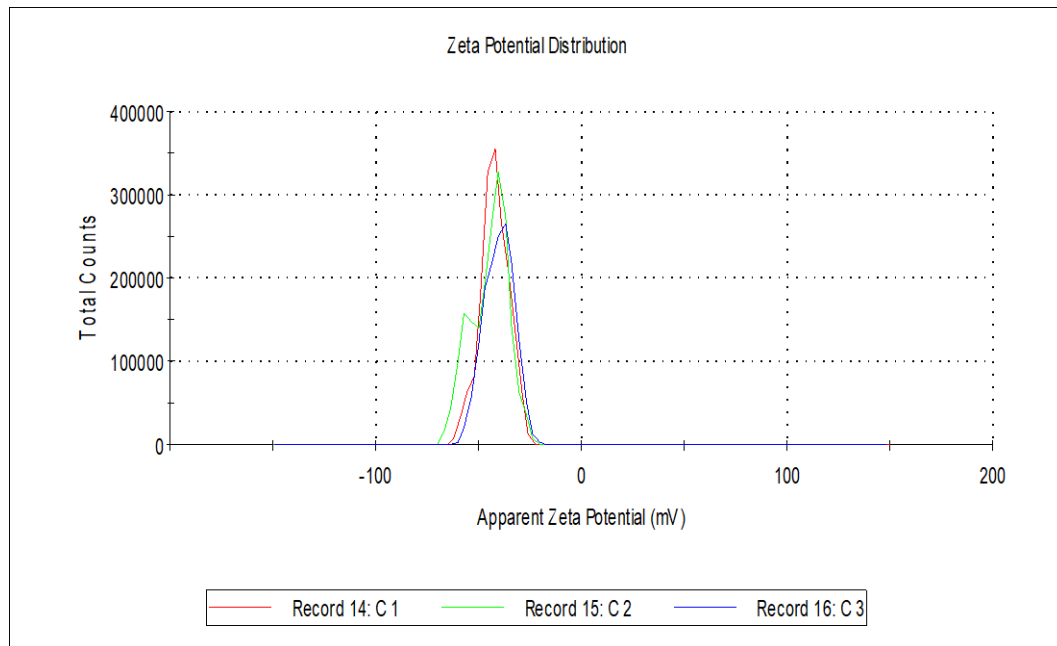


Fig 6.4. Zeta potential sample 3.5%

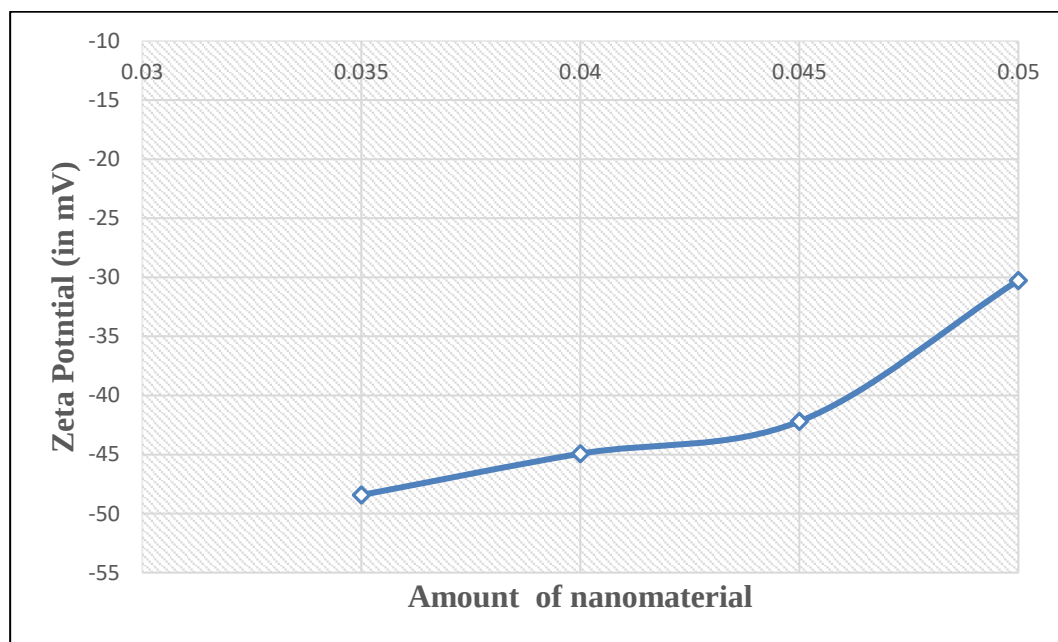


Fig 6.5. Zeta Potential versus amount of nanomaterial

The zeta potential of the samples ranging from 3.5% to 5% of the nanomaterial in the interval of 0.5% were analysed in the zetasizer. Each sample was being tested three times and then the average of the three was considered for the experiment. The zeta potential of the sample 5% of the nanomaterial was found to be -30.26mV. Thus the sample can be termed as moderately stable and was discarded as it can be critical for long use. This limits the use of nanomaterial in the PCM. Zeta potential value of all the other samples were well under the range and can be considered for further experimentation. It was seen that as we increase the

amount of nanomaterial, the value of zeta potential decreases. As the value zeta potential decreases, the nano-PCM will not resist aggregation and the attractive forces will form into small clumps or masses.

6.2 Latent Heat Degradation

Latent heat is defined as the heat or energy that is absorbed or released during a phase change of a substance. It could either be from a gas to a liquid or liquid to solid and vice versa. Latent heat is related to a heat property called enthalpy. However, an important point that we should consider regarding latent heat is that the temperature of the substance remains constant during a single phase. As far as the mechanism is concerned, latent heat is the work that is needed to overcome the attractive forces that hold molecules and atoms together in a substance.

Nanoparticle mixed with phase change material (PCM) modifies the crystallization ability. Mostly the crystallite size become smaller as during crystal growth the nanoparticle acts as impurity. This changes the heat of fusion.

Differential Scanning Calorimetry (DSC) is a thermoanalytical instrument which is used to measure the latent heat of Phase Change Materials (PCM). In this instrument, the difference for heat required to increase temperatures of a sample and a reference is measured as a function of temperature, which is usually programmed to change linearly as a function of time. In operation, an empty pair of aluminum or copper pan and lid is usually used as the reference and the sample is placed in another pair of pan and lid with the same material and weight as the reference, so that the heat flow to the reference is cancelled in software, and information of heat flow to the sample is collected.

Table 6.2. Latent heat degradation

Sr no.	Amount of nanomaterial	Latent heat (in KJ)
1	0%	170
2	3.5%	154.18
3	4%	143.25
4	4.5%	141.57

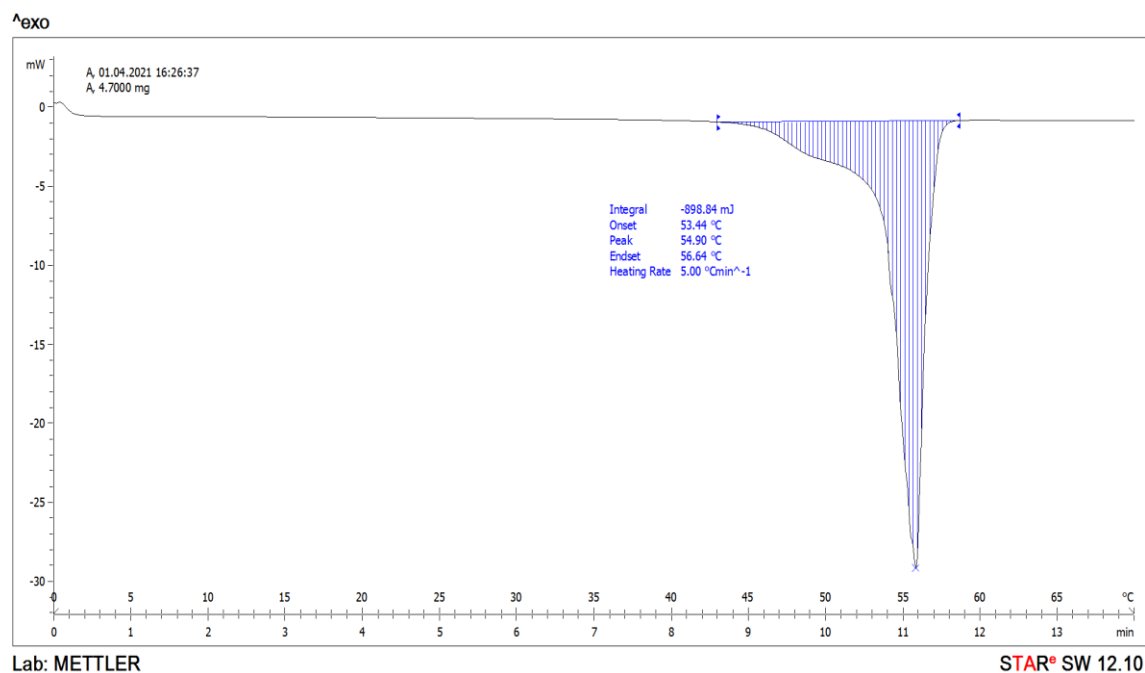


Fig 6.6 Latent Heat Sample 4.5%

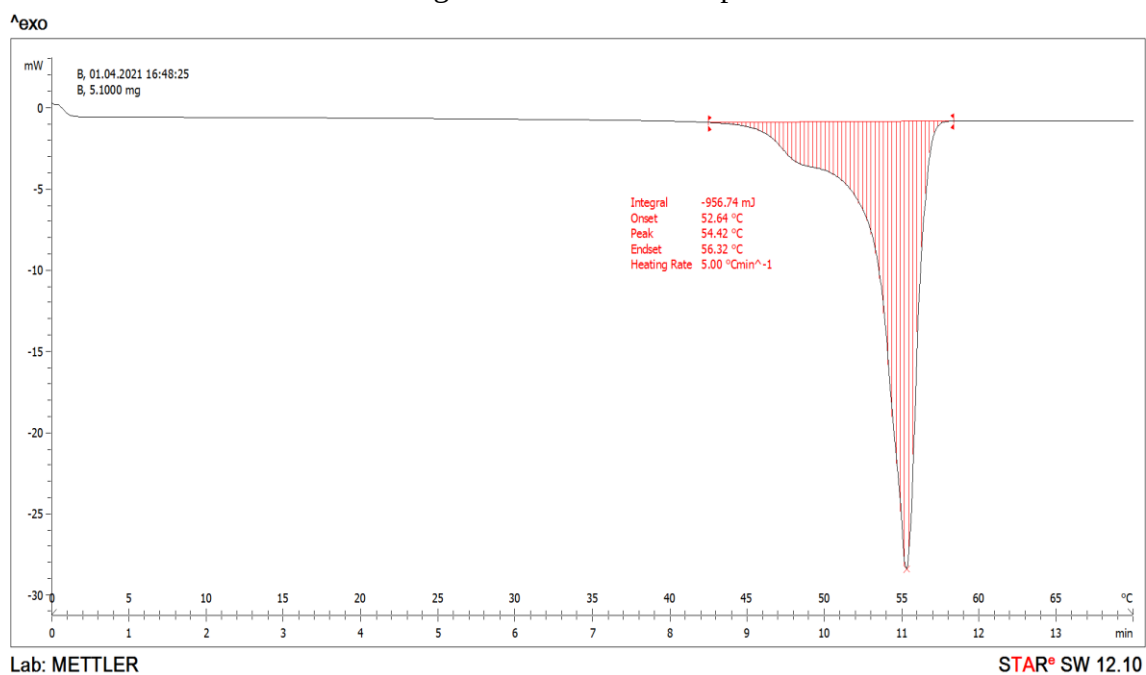


Fig 6.7. Latent Heat sample 4%

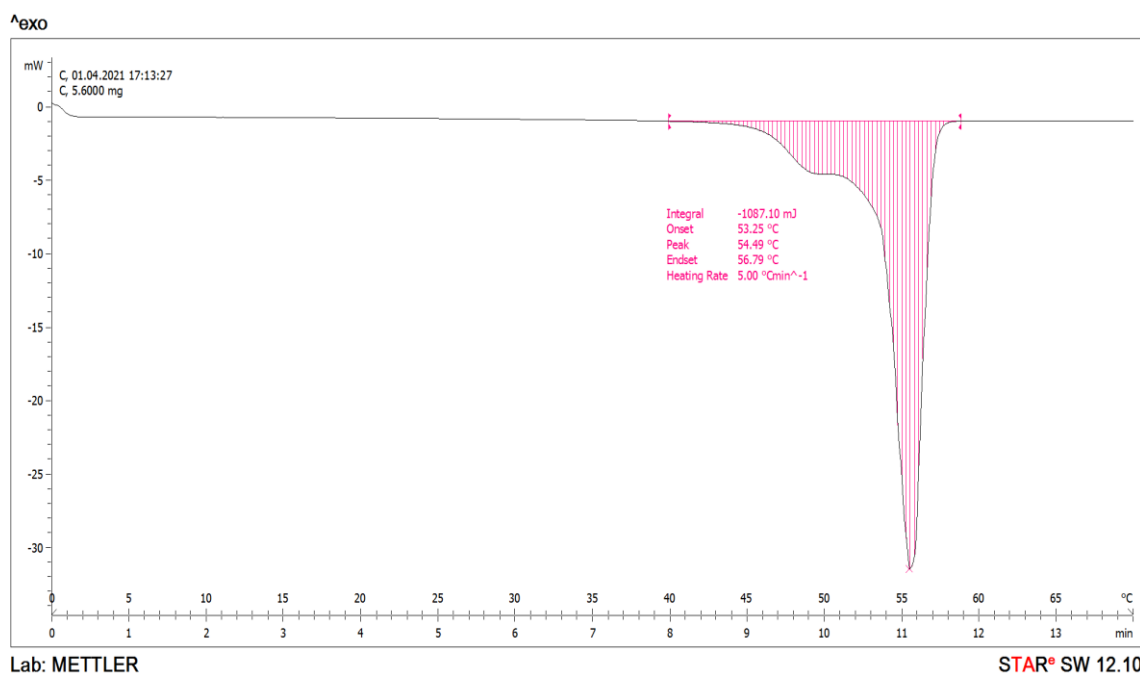


Fig 6.8. Latent Heat Sample 3.5%

The result of a DSC experiment is a curve of heat flux versus temperature or time, and the integration of phase transition peak area on the curve indicates the latent heat of the sample. The samples were tested in the DSC for 5 mg sample each and the heating rate was set as 5C/min. The onset was found to be 53C, peak at 54C and end set at 56c for all the samples tested. It was found out that as we increase the amount of nanomaterial, the latent heat of fusion decreases. The melting point of the PCM remains unaffected with the addition of nanomaterial.

Chapter 7

Conclusions and Future Scope

The results obtained from the test and experimentation were studied precisely to reach to the conclusions of the project. A keen observation and a clear study of the achieved graphs made us comprehend the working of the system and approach the conclusive interpretations and learn about the future study requirements which are mentioned in the further sections.

7.1 Conclusion

A detailed study of the results acquired was carried out which took the project to its final stage and the conclusions were made. The enrichment of normal PCM takes place due

to the addition of the nanomaterial which results in faster rate of heat transfer between the heat transfer fluid and the heat.

As the amount of nanomaterial in the PCM increases, the stability of the NePCM reduces. Hence, nanomaterial should not be used beyond a certain limit as it may affect its use for longer period. Another deciding factor which limits the use of nanomaterial is the degradation of latent heat. Hence, according to the application and its requirement the amount of nanomaterial must be selected.

7.2 Future Scope

After the successful conclusive study from the results obtained, there can be the interpretations regarding the future scope that must be followed for the further enhancement of this research. The current thermal energy storage system was made to work with selected PCM and nanomaterials which can be differed and different combinations of them with different proportions can be tried and analyzed to get the best results. Different method for the synthesis of NePCM can be used to get purer mixer.

The efficiency of the system can be studied by some passive methods with making changes in geometry by addition of fins to the internal pipe. Moreover, the conjunction of Fins as well as nanomaterials can also be examined.

The current study shows the fabrication of shell and tube heat exchanger which can be altered by taking some another type of heat exchanger for getting better performance. There is a lot of research going on worldwide in this field and has a lot of scope in its improvement with a view of approaching the best solution.

Appendices

Appendix A: Flow Sensor

The flow rate of the Heat Transfer Fluid (HTF) i.e. water was calculated post pumping from the motor using YF S201 Flow Sensor and Arduino Uno board. The results were displayed on a 16x2 LCD screen by maintaining the readings in Litres per Hour (LPH). The components used, the circuit diagram and the Arduino code used are mentioned below. Figure A.1. shows the schematic diagram of the Arduino circuit for the flow sensor.

Components Used:

- YF S201 Flow Sensor
- Arduino Uno Board
- 16 x 2 LCD display
- Breadboard
- Connecting Wires

Circuit Diagram:

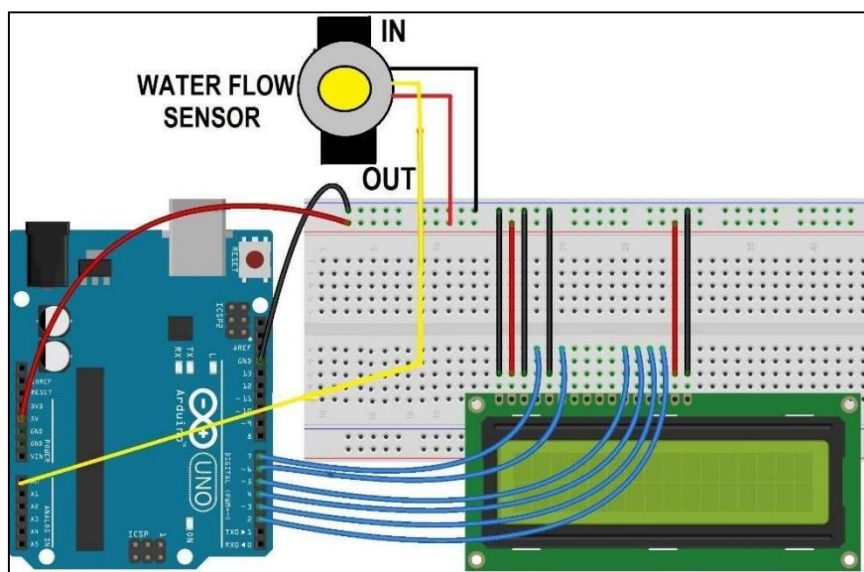


Fig A.1. Arduino Circuit diagram for Flow Sensor and LCD

Arduino Uno code:

```
#include <LiquidCrystal.h>
LiquidCrystal lcd(7, 6, 5, 4, 3, 2); int X;
int Y;
float TIME = 0;
float FREQUENCY = 0;
float WATER = 0;
float TOTAL = 0;
float LS = 0;
const int input = A0;
void setup()
{
  Serial.begin(9600);
  lcd.begin(16, 2);
  lcd.clear();
  lcd.setCursor(0,0);
  lcd.print("Water Flow Meter");
  lcd.setCursor(0,1);
  lcd.print("*****");
  delay(2000);
  pinMode(input,INPUT);
}
void loop()
{
  X = pulseIn(input, HIGH); Y
  = pulseIn(input, LOW);
  TIME = X + Y;
  FREQUENCY = 1000000/TIME;
  WATER = (FREQUENCY/7.5)*60; LS
  = WATER/60;
  if(FREQUENCY >= 0)
  {
    if(isinf(FREQUENCY))
    {
      lcd.clear();
      lcd.setCursor(0,0);
      lcd.print("FLOW RATE");
      lcd.setCursor(0,1);
      lcd.print("0.00");
      lcd.print("LPH");
    }
    else
    {
      lcd.clear();
      lcd.setCursor(0,0);
      lcd.print("FLOW RATE");
      lcd.setCursor(0,1);
      lcd.print(WATER);
      lcd.print("LPH");
    }
  }
  delay(1000);
}
```

Appendix B: Data Acquisition System

K type thermocouples were used for measuring the temperatures at various locations as mentioned earlier. Co-operatively, Data Acquisition system was used that measures an electrical or physical phenomenon such as current, voltage, temperature, pressure, etc. with a use of computer. A DAQ system consists of sensors, DAQ measurement hardware and a computer with programmable software. Compared to the traditional measurement system, these computer-based DAQ systems exploit the productivity, connecting capabilities, processing power, display of the parameters to be measured, etc. providing a more flexible, more accurate and efficient measurement solution. Figure B.1. shows the block diagram for a general Data Acquisition System.

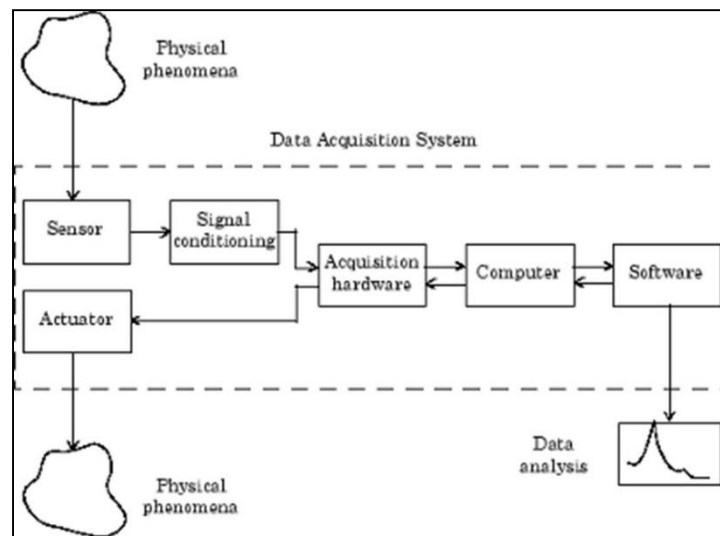


Fig B.1. Block Diagram for Data Acquisition System (DAQ)

For the current project setup, total number of 10 thermocouples were used to measure temperatures at different locations simultaneously. Hence DAQ system manufactured by National Instruments modelled as NI-DAQ USB 6211 was used for acquiring the temperature readings. The NI-DAQ USB-6211 is a multifunction DAQ device which offers analog I/O, digital I/O, and two 32-bit counters. The device provides an onboard amplifier designed for fast settling times at high scanning rates.

The device is ideal for test, control, and design applications including portable data logging, field monitoring, embedded OEM, in-vehicle data acquisition, and academic. The USB-6211 features a lightweight mechanical enclosure and is USB powered for easy portability. The actual product image of the DAQ hardware is shown in the figure B.2.

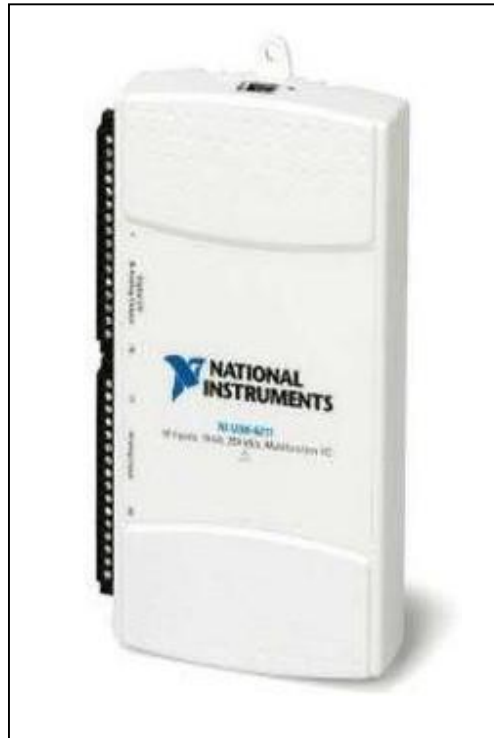


Fig B.2. NI-DAQ USB 6211

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